How Architecture Topology Shapes Normalization-Free Network Effectiveness

Anonymous Author(s)
Affiliation
Address
`email`

Abstract

Dynamic Tanh (DyT) promises to replace LayerNorm with a simple learnable nonlinearity, and delivers on standard Vision Transformers. But modern vision architectures have diversified far beyond vanilla attention into state-space models (SSMs), hybrids, and deep variants with information-flow topologies that reshape how normalization errors propagate. We test DyT across ten architectures spanning three topologies (open-loop attention, closed-loop SSM, hybrid) on CIFAR-100 and Tiny-ImageNet. The outcome is decisively architecture-dependent: DyT boosts DeiT-S by $+6.90$ pp yet degrades Swin-T by -17.18 pp, a 24 pp spread between two attention-based models that differ mainly in locality. Closed-loop SSMs split along scan direction; the hybrid RWKV-T degrades even though it matches the depth and width of a benefiting architecture. A three-factor framework (depth–width ratio, multi-layer perceptron (MLP) availability, topology) accounts for six of ten outcomes. Normalization replacement is not a drop-in operation: the architectural role of normalization must be understood before it can be safely removed.

1 Introduction

LayerNorm (LN) [Ba et al., 2016] is ubiquitous in modern Transformers [Vaswani et al., 2017], but introduces per-token statistical computations that complicate hardware-efficient inference. Dynamic Tanh (DyT) [Zhu et al., 2025] offers an appealing alternative: replace LayerNorm with $\gamma \cdot \tanh(\alpha\, x) + \beta$, a fixed-form nonlinearity with no running statistics that matches performance on Vision Transformers [Dosovitskiy et al., 2021] and language models. If a learnable $\tanh$ suffices where LayerNorm was thought necessary, normalization may be an architectural habit rather than a structural requirement.

DyT's promise, however, has been tested only on open-loop attention architectures [Zhu et al., 2025]. Vision architectures now span fundamentally different *information-flow topologies*: state-space models (SSMs) like Mamba/Vim [Gu and Dao, 2024, Zhu et al., 2024] introduce closed-loop recurrence where normalization errors compound through state updates rather than dissipating independently per layer; hybrids like Vision-RWKV [Duan et al., 2024, Peng et al., 2023] interleave linear attention with specialized normalization sites (e.g., `key_norm`) that serve structural roles beyond standardization. Whether DyT transfers across these topological boundaries is an open question, and one with practical stakes: practitioners need to know where DyT breaks, not just where it works.

We address this gap through controlled experiments on CIFAR-100 and Tiny-ImageNet across ten architectures: DeiT-S/T (open-loop) [Touvron et al., 2021a], Vim-T and PlainMamba-T (closed-loop) [Zhu et al., 2024, Yang et al., 2024], RWKV-T (hybrid) [Peng et al., 2023], with probes on

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CaiT-XXS24 [Touvron et al., 2021b], Sequencer2D-S [Tatsunami and Taki, 2022], RMT-T [Fan et al., 2024], ConvNeXt-T, and Swin-T. Our findings:

1. **DyT effectiveness is decisively architecture-dependent.** On CIFAR-100, DyT boosts DeiT-S by $+6.90$ pp but degrades Swin-T by -17.18 pp, a 24 pp spread between two attention-based models that differ mainly in locality. Only global standard self-attention benefits; windowed, deep, convolutional, and retention-based variants all degrade or collapse (Table 1). This sharp divergence within a single topology raises the question: what structural properties determine the outcome?

2. **Three interacting factors account for DyT compatibility.** A framework based on depth–width ratio, multi-layer perceptron (MLP) compensatory pathways, and information-flow topology accounts for six of ten architectures (Table 2); α-sensitivity sweeps and RMSNorm controls confirm the pattern is topology-intrinsic (Appendix N, U). Among closed-loop SSMs, scan direction emerges as an independent axis: Vim-T ($+3.45$ pp) and PlainMamba-T (-0.41 pp) diverge despite similar scale. Cross-dataset validation on Tiny-ImageNet replicates the directional pattern (Appendix T). Yet the framework has a boundary: at higher resolution, a systematic failure mode emerges.

3. **Resolution scaling exposes an α-deadlock, and hybrid degradation reveals site-specific roles.** On ImageNet-100 at 224×224, all tested α values converge 18–23 pp below LN, consistent with tanh saturation dynamics [Glorot and Bengio, 2010, Chen et al., 2025] (§K). RWKV-T's degradation further decomposes into structural norms ($\sim$65%) and `key_norm` ($\sim$35%). This decomposition demonstrates that even partial replacement requires understanding of each normalization site's functional role.

2 Related Work

2.1 Normalization in Deep Learning

Batch Normalization (BN) [Ioffe and Szegedy, 2015] normalizes activations across the mini-batch dimension but depends on batch statistics. This dependence makes BN ill-suited for small batches or variable-length sequences. LayerNorm (LN) [Ba et al., 2016] normalizes across features within each sample and has become the default in Transformers [Vaswani et al., 2017]. Variants include GroupNorm [Wu and He, 2018] for convolutional architectures and RMSNorm [Zhang and Sennrich, 2019], which removes mean-centering for efficiency. Theoretically, Santurkar et al. [2018] show that normalization smooths the loss landscape rather than reducing internal covariate shift. This finding provides a principled basis for understanding why normalization removal may destabilize certain architectures.

Beyond activation normalization, Query-Key Normalization [Henry et al., 2020] stabilizes attention logits by constraining query and key magnitudes. The coexistence of these functionally distinct normalization roles complicates uniform replacement.

Pre-norm placement yields more stable gradients than post-norm [Xiong et al., 2020], while DeepNet [Wang et al., 2024] stabilizes Post-LN via residual scaling. Both findings underscore that normalization effectiveness is architecture-dependent, a theme central to our investigation.

2.2 Normalization-Free Networks

In the convolutional setting, Fixup [Zhang et al., 2019] replaces BN via careful initialization in ResNets [He et al., 2016], and NFNets [Brock et al., 2021] combine signal propagation analysis with Adaptive Gradient Clipping. Both are tailored to CNNs and do not transfer to Transformers, where unbounded attention scores and residual stream activations pose distinct challenges. ReZero [Bachlechner et al., 2021] bridges this gap by initializing residual branch weights to zero in Transformers. This approach enables norm-free training through signal propagation control rather than activation normalization.

Dynamic Tanh (DyT) [Zhu et al., 2025] takes a different approach: replacing LN with a learnable bounded nonlinearity $\gamma \cdot \tanh(\alpha \cdot x) + \beta$ as a *post-hoc substitution* that keeps training procedures unchanged. The tanh saturation connects to classical vanishing gradient findings [Glorot and Bengio, 2010]. Concurrently, Chen et al. [2025] propose Derf, an alternative point-wise replacement based on the Gaussian error function that improves upon DyT through better generalization properties. Stollenwerk [2025] further show that DyT approximates RMSNorm and derive an exact point-wise equivalent. This result provides theoretical grounding for replacement fidelity. Independently, nGPT [Loshchilov et al., 2024] constrains all representations to the unit hypersphere and thereby eliminates normalization entirely through geometric constraints rather than activation replacement. All three approaches were validated primarily on global self-attention architectures. Applicability of these three approaches to other topologies remains unexplored.

These efforts focus on *whether* normalization can be removed; our work asks *when*: under what topology conditions a representative LN replacement succeeds or fails.

2.3 Vision Architecture Topologies

Modern vision architectures span a spectrum of information-flow topologies. ViT [Dosovitskiy et al., 2021] and DeiT [Touvron et al., 2021a] process tokens through stacked global self-attention in a purely feedforward (*open-loop*) manner. CaiT [Touvron et al., 2021b] modifies this pattern by separating self-attention and class-attention into distinct stages (a *class-attention* topology) and introduces additional normalization sites at the class-token interaction boundary. Swin [Liu et al., 2021] restricts attention to local windows with hierarchical pooling, and ConvNeXt [Liu et al., 2022] modernizes CNNs with large-kernel depthwise convolutions and LN. Both adopt a *hierarchical* topology that produces multi-resolution feature maps rather than a columnar token sequence.

State-space models introduce recurrent dynamics. S4 [Gu et al., 2022] provides the continuous-time foundation; Mamba [Gu and Dao, 2024] adds input-dependent selection with a hidden state updated recurrently. Vision Mamba (Vim) [Zhu et al., 2024] adapts this to image patches with bidirectional scanning and creates a *closed-loop* feedback path. PlainMamba [Yang et al., 2024] further simplifies the architecture by removing MLP blocks entirely; normalization thus becomes the sole inter-layer mechanism that regulates signal magnitude before recurrent state transitions. Sequencer2D [Tatsunami and Taki, 2022] applies bidirectional LSTMs along both spatial axes. xLSTM [Beck et al., 2024], Mamba-2 [Dao and Gu, 2024], and VMamba [Liu et al., 2024] provide additional closed-loop designs.

Hybrid architectures lie between these extremes. Vision-RWKV [Duan et al., 2024], based on RWKV [Peng et al., 2023], interleaves linear attention with spatial convolution priors and employs functional normalization (`key_norm`) beyond generic activation stabilization; RWKV-5/6 [Peng et al., 2024] extend the RWKV framework with matrix-valued states. RetNet [Sun et al., 2023] and RMT [Fan et al., 2024] similarly blend parallel attention with recurrent computation.

This topology distinction is central to our analysis: prior normalization replacement work treats architectures as interchangeable targets, whereas our results reveal information-flow topology as a primary factor determining replacement feasibility.

2.4 Architecture-Dependent Normalization

The functional role of normalization varies across architecture families. In open-loop Transformers, LN stabilizes per-token activations in the residual stream. In CaiT [Touvron et al., 2021b], additional normalization at the class-attention boundary mediates information flow between self-attention and class-attention stages, a structural role absent from standard ViTs. In closed-loop SSMs [Gu and Dao, 2024, Zhu et al., 2024], LN additionally constrains recurrent state dynamics; PlainMamba [Yang et al., 2024] makes this coupling especially tight by removing MLP blocks, so normalization directly gates state transitions without an intermediate nonlinear bottleneck. In hybrids such as Vision-RWKV [Duan et al., 2024], certain normalization layers (e.g., `key_norm`) serve structural roles [Henry et al., 2020] that generic replacements cannot replicate. Hierarchical designs such as Swin [Liu et al., 2021] and ConvNeXt [Liu et al., 2022] add further complexity: LN must stabilize activations across resolution stages where feature statistics change abruptly at downsampling boundaries. ConvNeXt

V2 [Woo et al., 2023] further illustrates this architecture-specificity by introducing Global Response Normalization (GRN), a channel-wise normalization tailored to convolutional feature maps that standard LN or its replacements cannot subsume.

Within the SSM family, Feng et al. [2025] systematically study normalization type, position, and combinations within the Mamba block, finding that post-SSM normalization and heterogeneous norm pairings improve training stability and downstream performance. Independently, Lahoti et al. [2026] introduce BC normalization (BCNorm) in Mamba-3, applying RMSNorm after the B and C projections—analogous to QKNorm in Transformers—which stabilizes large-scale training and permits removal of the post-gate RMSNorm layer. Both studies optimize normalization design *within* a single architecture family; our work instead asks *across* families whether architectural topology determines if normalization can be removed entirely.

No prior work has systematically compared normalization replacement across architectures with distinct information-flow topologies.

2.5 DyT Across Architectures

Zhu et al. [2025] and Chen et al. [2025] evaluated DyT on ViT, DiT, and LLMs, all global self-attention architectures. Whether DyT transfers to closed-loop, hierarchical, or hybrid architectures has not been investigated.

We fill this gap by evaluating DyT across ten vision architectures spanning open-loop, closed-loop, class-attention, hierarchical, and hybrid topologies on CIFAR-100 and Tiny-ImageNet. The results provide the first systematic evidence that architecture topology is a primary determinant of whether LN can be safely replaced. Only global standard self-attention architectures (DeiT-S/T) consistently benefit; hierarchical, class-attention, and hybrid designs degrade or collapse, while closed-loop SSMs exhibit subtype-dependent outcomes: bidirectional scanning (Vim-T) gains whereas unidirectional designs with directional bias (PlainMamba) do not.

3 Method

3.1 Architecture Taxonomy

We select four architectures that span three information-flow topologies and share the *pre-norm residual pattern* with token-wise normalization, common to standard Transformers, SSMs, and linear attention variants. We exclude hierarchical CNNs, where normalization manages cross-stage scale transitions. All models process 32×32 images patchified into token sequences.

Open-loop: DeiT-S and DeiT-T. DeiT [Touvron et al., 2021a] computes multi-head self-attention over all token pairs followed by a feed-forward network, with no recurrent state (*open-loop*). We include DeiT-S ($\sim$22M) and DeiT-T ($\sim$5.4M) to disentangle topology from capacity effects.

Closed-loop: Vim-T. Vision Mamba (Vim) [Zhu et al., 2024] applies Mamba's [Gu and Dao, 2024] Selective State Space Model to vision and maintains a hidden state updated recurrently: $h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t$, $y_t = C_t h_t$. This recurrence constitutes a *closed-loop* feedback path ($\sim$22M parameters).

Hybrid: RWKV-T. Vision-RWKV [Duan et al., 2024], based on RWKV [Peng et al., 2023], combines convolutional priors (q_shift) with bidirectional WKV attention. Its distinctive `key_norm` bounds key magnitudes before attention computation. This *hybrid* architecture is neither purely feedforward nor recurrent ($\sim$5.8M parameters).

We further extend evaluation to six additional architectures (PlainMamba-T, CaiT-XXS24, ConvNeXt-T, Swin-T, RMT-T, Sequencer2D-S) as probes; details appear in §4 and the Appendix.

3.2 DyT Replacement Protocol

DyT [Zhu et al., 2025] replaces LayerNorm with a learnable bounded nonlinearity:

$$\mathrm{DyT}(x) = \gamma \cdot \tanh(\alpha\, x) + \beta, \tag{1}$$

where α controls saturation and γ, β are affine parameters. Unlike LN, DyT applies a pointwise nonlinearity without per-token statistics.

Universal replacement. We traverse the full module tree of each architecture and substitute every LayerNorm with DyT (initialized $\alpha = 0.5$). This protocol applies uniformly across all ten architectures ($< 0.1\%$ parameter change).

Key-norm ablation. RWKV-T's `key_norm` serves a structural role distinct from standard pre-/post-block norms. We introduce selective exclusion: three configurations (full LN baseline, full DyT, and DyT with `key_norm` preserved as LN) isolate the key normalization contribution.

3.3 Training Setup

Datasets. Primary experiments use CIFAR-100 (100 classes, 50K training images, 32×32, standard test split); the rationale for small-scale evaluation is discussed in Appendix L. Cross-dataset validation uses Tiny-ImageNet (200 classes, 100K images, 64×64 resized to 32×32).

Augmentation and regularization. We apply mixup [Zhang et al., 2018] ($\alpha = 0.8$), cutmix [Yun et al., 2019] ($\alpha = 1.0$), label smoothing [Szegedy et al., 2016] ($\varepsilon = 0.1$), and stochastic depth [Huang et al., 2016] (rate $= 0.1$), all held constant across architectures.

Optimization. We train all models from scratch with AdamW [Loshchilov and Hutter, 2019] (lr $= 10^{-3}$, weight decay $= 0.05$), cosine schedule [Loshchilov and Hutter, 2017] without warmup, 100 epochs, batch size 128, FP32, patch size 4 (64 tokens), on a single NVIDIA RTX 5090. No architecture-specific tuning is applied.

Evaluation and controls. We report best top-1 test accuracy averaged over three seeds (42, 123, 0). Two parameter-matched pairs, DeiT-S/Vim-T ($\sim$22M) and DeiT-T/RWKV-T ($\sim$5.5M), isolate topology from scale.

Reproducibility. Implementation based on PyTorch and timm. Code available upon publication.

4 Results

4.1 Architecture Topology and DyT Effectiveness at Low Resolution

DyT effectiveness partitions sharply by architecture topology. Table 1 compares LayerNorm and DyT across ten architectures on CIFAR-100 (100 epochs; 3-seed mean for eight models, single-seed for two probes marked †).

Three groups emerge (Figure 1): positive ($\Delta \geq +3$ pp: DeiT-S/T, Vim-T), near-zero (PlainMamba-T), and negative or collapsed (remaining six). Global self-attention at moderate depth yields the largest gains (DeiT-S at $+6.90$ pp), while deeper or windowed variants degrade substantially. Swin-T's deficit is structural rather than initialization-related (Table 19). Among closed-loop SSMs, Vim-T gains $+3.45$ pp while PlainMamba-T is near-zero (Appendix F); the two differ in scan direction and MLP availability (Appendix R). The hybrid RWKV-T drops -2.17 pp, decomposable into structural ($\sim$65%) and `key_norm` ($\sim$35%) components (Appendix Q). ConvNeXt-T's high LN variance (σ=2.45 pp) complicates direct comparison (Appendix E).

Notably, DyT improves corruption robustness even where clean accuracy drops: on CIFAR-100-C, RWKV-T's mean corruption accuracy rises $+2.73$ pp under DyT despite the -2.17 pp clean

Table 1: LayerNorm vs. DyT on CIFAR-100 (%, 3-seed mean±std; †single seed). Δ: DyT−LN (pp). Best α from sweep (§4.3); "—" = not swept. ‡Non-standard hyperparameters. §High LN variance (σ=2.45 pp). ^{w}5-epoch warmup (LN/DyT: no warmup).

Architecture	Topology	Params	LN Acc (%)	RMSNorm Acc (%)	DyT Acc (%)	Δ (pp)	Best α
DeiT-S	Open-loop (attn)	22M	58.0 ± 1.6	60.7 ± 0.3^{w}	64.9 ± 0.6	**+6.90**	0.50
DeiT-T	Open-loop (attn)	5.4M	56.9 ± 0.1	—	60.8 ± 0.6	+3.91	—
Vim-T	Closed-loop (SSM)	22M	61.1 ± 0.6	—	64.5 ± 0.8	+3.45	0.50
PlainMamba-T	Closed-loop (SSM)	12.1M	74.2 ± 0.1	—	73.8 ± 0.2	−0.41	—
RWKV-T	Hybrid	5.8M	74.6 ± 0.2	72.4 ± 0.2	72.5 ± 0.2	−2.17	1.00
CaiT-XXS24	Open-loop (deep attn)	11.6M	68.9 ± 0.1	—	56.6 ± 0.4	−12.34	2.00
ConvNeXt-T§	Open-loop (CNN)	27.9M	55.1 ± 2.5	—	43.0 ± 0.7	−12.11	—
Swin-T‡	Open-loop (windowed attn)	27.6M	64.0 ± 0.1	—	46.8 ± 0.3	−17.18	1.0
RMT-T†	Open-loop (retention)	13.4M	56.4	—	collapse (30.1)	—	—
Sequencer2D-S†	Closed-loop (BiLSTM)	27M	61.1	—	collapse (5.2)	—	—

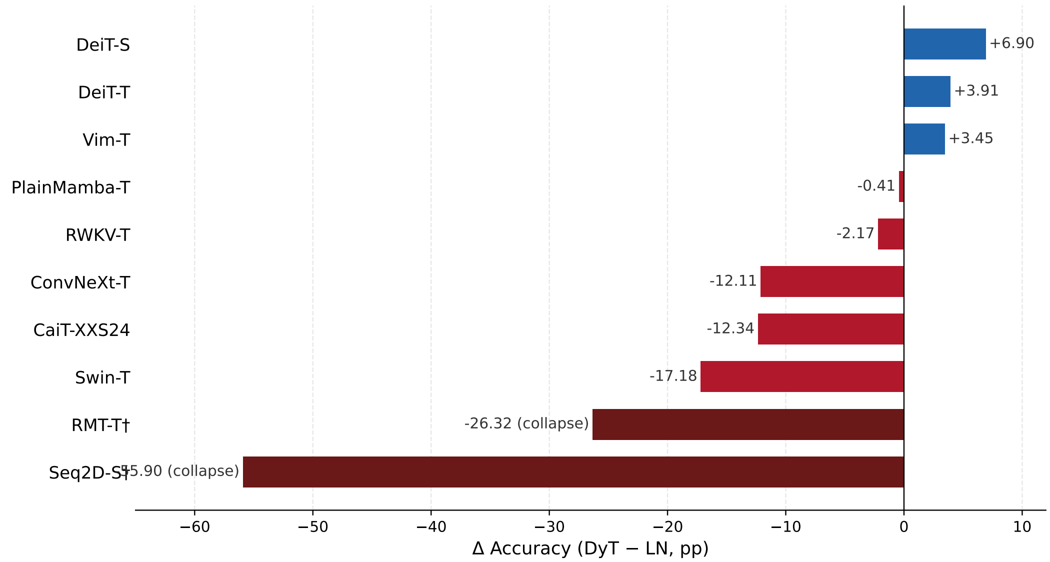

Figure 1: Per-model Δ accuracy (DyT − LayerNorm) on CIFAR-100 across ten models. Positive values (blue) indicate DyT improvement; negative bars (red) indicate degradation. Swin-T (−17.18 pp) exhibits the largest deficit among converging architectures.

deficit. This accuracy gain points to an implicit regularization effect beneficial for out-of-distribution generalization (Appendix C; extended discussion in Appendix I). Convergence speed and cross-dataset analyses appear in Appendices A and T.

4.2 Disentangling Topology from Scale

Can parameter scale alone explain the divergent outcomes? Two parameter-matched pairs isolate topology from capacity: at ~22M, DeiT-S (+6.90 pp) and Vim-T (+3.45 pp) both benefit, though topology modulates the magnitude; at ~5.5M, DeiT-T (+3.91 pp) and RWKV-T (−2.17 pp) show a sign reversal that scale cannot explain. Formal confound analyses controlling for parameter count and baseline accuracy confirm that topology remains the dominant correlate (Appendices G and H).

4.3 Sensitivity to Initialization Scale (α)

How sensitive are the observed accuracy gaps to initialization scale? The α-sensitivity sweep (Appendix N, Table 19) reveals topology-dependent stability: Vim-T converges only at α=0.50; DeiT-S at $\alpha \leq 0.50$; RWKV-T at all tested values and largely closes the gap at α=1.0. Swin-T and CaiT-XXS24 converge broadly but remain $\geq$6.6 pp below LN, confirming structural deficits. The

Table 2: Three-factor analysis of DyT effectiveness. Each row shows one of the six architectures accounted for by the framework. Controlled pairs (matched depth or width) isolate individual factors.

Architecture	L	d	Topology	MLP	Δ (pp)
DeiT-S	12	384	Open (global SA)	✓	$+6.90$
DeiT-T	12	192	Open (global SA)	✓	$+3.91$
CaiT-XXS24	26	192	Open (class-attn)	✓	-12.34
Vim-T	10	384	Closed (bi-scan)	✓	$+3.45$
PlainMamba-T	12	384	Closed (uni-scan)	—	-0.41
RWKV-T	12	192	Hybrid	✓	-2.17

need for per-architecture α tuning is itself topology-dependent and complements the direction of DyT's effect.

RMSNorm controls confirm topology-intrinsic normalization dependence rather than DyT-specific artifacts (Appendix U).

5 Discussion

Three interacting factors emerge from the pattern in Table 1: depth–width ratio, MLP compensatory pathways, and topology-specific propagation. Together, these account for six of ten architectures (Table 2, Figure 2); the remaining four require architecture-specific considerations (extended to all ten architectures in Appendix S). Width buffers tanh compression (DeiT-S $+6.90$ vs. DeiT-T $+3.91$ pp at 12L); depth amplifies this compression (DeiT-T vs. CaiT-XXS24: sign reversal at $d{=}192$). MLP ablations (Appendix R) show MLP is necessary but insufficient: removal reverses Vim-T's Δ ($\sim$12 pp swing), while grafting MLP onto PlainMamba-T does not help; this outcome implicates scan direction as an independent factor (Appendix W.1). The framework is post-hoc and correlational.

Mechanism integration. These three factors converge on a unified picture: DyT's tanh nonlinearity interacts differently with each architecture's information flow. Learned α correlates with saturation and rank compression (Appendix V), and RMSNorm controls confirm the pattern is normalization-intrinsic rather than DyT-specific (Appendices U and W.3). Selective preservation of functional norms (e.g., RWKV-T `key_norm`) recovers $\sim$35% of degradation. This recovery motivates topology-aware partial replacement (Appendices Q, W.2). Cross-dataset analysis confirms these patterns generalize beyond CIFAR-100 (Appendix W.4).

We note that RWKV-T replaces 50 normalization sites versus 25 in DeiT-S; however, PlainMamba-T (13 sites, $\Delta{\approx}0$) suggests that replacement count alone does not determine the outcome.

5.1 Limitations

Our experiments use 32×32 resolution (Appendix L); ImageNet-100 probes (§K) show DyT degrades 18–23 pp below LN at 224×224. The 2–3 seed protocol establishes directional trends but precludes formal significance testing; RMT-T and Sequencer2D-S are single-seed. The framework is post-hoc and correlational; positive results are concentrated in the DeiT family. DyT also incurs modest memory overhead (+8.7–23.4%) with variable throughput impact (Appendices B and J).

6 Conclusion

Our cross-topology evaluation reveals that DyT effectiveness is not a property of DyT itself, but of the architecture it is applied to. The primary determinant is information-flow topology: the same replacement that improves global self-attention degrades windowed, deep, and hybrid variants. Closed-loop SSMs split along finer-grained structural axes such as scan direction.

A three-factor framework (depth–width ratio, MLP compensatory pathways, and topology-specific propagation) organizes these outcomes into a coherent pattern and accounts for six of ten architectures.

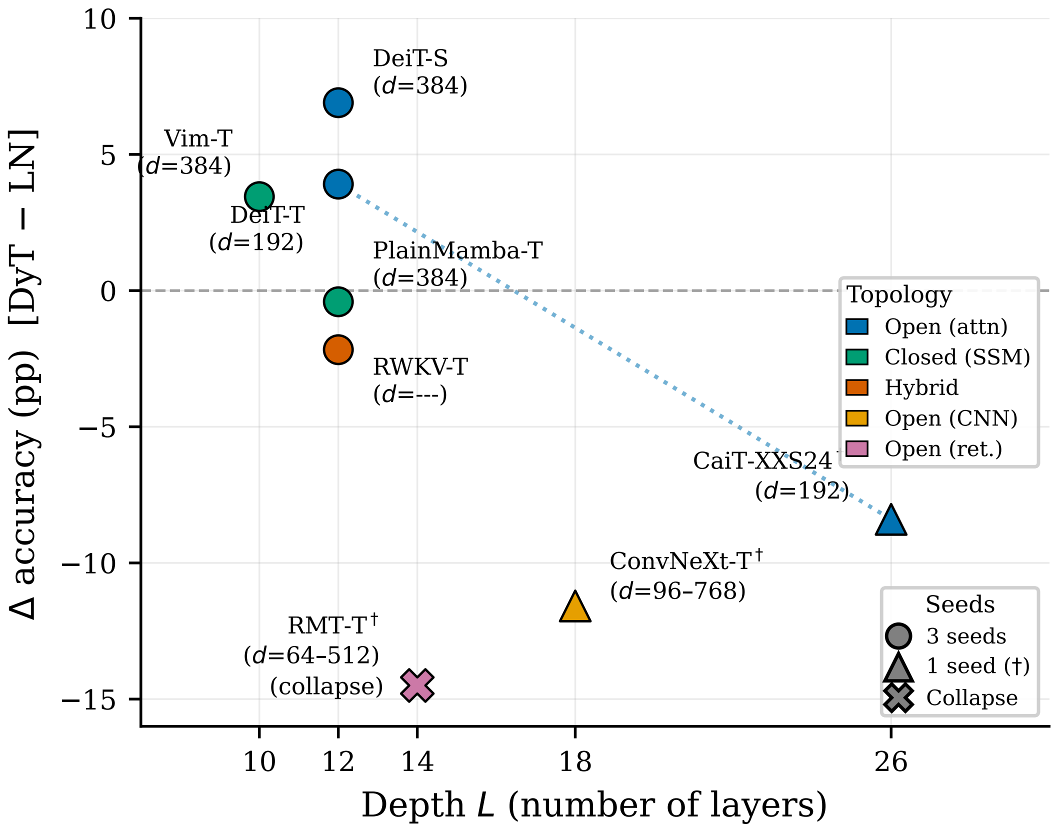

Figure 2: Depth vs. DyT Δ accuracy on CIFAR-100. Color encodes topology; marker shape encodes seed count. The dotted line connects d=192 open-loop points (DeiT-T $\rightarrow$ CaiT-XXS24). RMT-T (collapse) is clipped.

The framework is post-hoc and correlational, but it converts an unpredictable per-model inventory into a structured analysis with empirically grounded hypotheses.

The practical implication is direct: replacing LayerNorm with DyT is not a drop-in operation. Each normalization site may serve a different functional role: standardization, gradient conditioning, or structural gating. Substitutability depends on which role dominates. Topology-aware selective replacement offers a principled path forward: preserve functional norms and substitute only redundant ones.

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A Convergence Speed Analysis

Table 3: Convergence speed and best validation accuracy on CIFAR-100 (100 epochs, mean±std across seeds). Epoch→50% / 90%: first epoch reaching 50% / 90% of that run's best accuracy. Best Epoch: epoch at which best accuracy is achieved.

Architecture	Norm	Seeds	Best Acc (%)	Epoch→50%	Epoch→90%	Best Epoch
DeiT-S	LN	3	58.0 ± 1.6	19.0 ± 2.6	53.0 ± 5.3	91.0 ± 0.0
DeiT-S	DyT	3	64.9 ± 0.6	10.0 ± 1.0	47.7 ± 1.2	92.0 ± 3.6
DeiT-T	LN	3	56.9 ± 0.1	14.7 ± 1.5	55.7 ± 3.2	94.0 ± 1.0
DeiT-T	DyT	3	60.8 ± 0.6	12.0 ± 1.0	50.0 ± 3.5	95.0 ± 6.2
Vim-T	LN	3	61.1 ± 0.6	6.7 ± 1.2	38.7 ± 1.5	91.0 ± 5.3
Vim-T	DyT	3	64.5 ± 0.8	7.3 ± 0.6	35.7 ± 1.5	91.7 ± 2.5
RWKV-T	LN	3	74.6 ± 0.2	5.0 ± 0.0	31.0 ± 0.0	92.3 ± 3.1
RWKV-T	DyT	3	72.5 ± 0.2	7.0 ± 0.0	36.0 ± 1.0	93.3 ± 4.5

Table 3 jointly reports convergence speed and peak accuracy. DyT improves best accuracy for both open-loop attention architectures and the closed-loop Vim-T, but *degrades* it for the hybrid RWKV-T, consistent with the main results (Table 1).

Despite DyT's faster early convergence—DeiT-S with DyT reaches 50% accuracy nearly twice as fast as LN (10.0 ± 1.0 vs 19.0 ± 2.6 epochs) and the 90% threshold is also reached earlier (47.7 ± 1.2 vs 53.0 ± 5.3)—the epoch at which best accuracy is achieved (Best Epoch) is uniformly late across all conditions (∼89–99), showing no systematic difference between normalizations. This indicates that DyT accelerates the *trajectory* through the loss landscape without shifting the point of peak generalization, which is largely governed by the cosine learning rate schedule.

For RWKV-T, DyT slows convergence at both thresholds (7.0 vs 5.0 for 50%; 36.0 vs 31.0 for 90%) while also lowering peak accuracy, reinforcing the finding that the effect of replacing LayerNorm with DyT is topology-dependent, with hybrid architectures showing degradation.

B Throughput Benchmark

Table 4: Inference throughput and latency on a single NVIDIA RTX 5090 GPU. All models are evaluated with batch size 128, 32×32 input resolution, FP32 precision. Each measurement averages 500 timed iterations after 100 warmup iterations. Δ Throughput reports the relative change when replacing LN with DyT.

Architecture	Norm	Throughput (img/s)	Latency (ms/batch)	Δ Throughput (%)
DeiT-T	LN	34 760	3.68	—
DeiT-T	DyT	33 349	3.84	−4.1
DeiT-S	LN	12 186	10.50	—
DeiT-S	DyT	11 713	10.93	−3.9
RWKV-T	LN	13 817	9.26	—
RWKV-T	DyT	11 745	10.90	−15.0
Vim-T	LN	9 522	13.44	—
Vim-T	DyT	9 277	13.80	−2.6

DyT introduces a modest throughput reduction across all architectures compared to LN. The open-loop DeiT variants and the closed-loop Vim-T show relatively small slowdowns (2.6–4.1%), whereas RWKV-T exhibits a substantially larger degradation (−15.0%). The disproportionate overhead in RWKV-T likely stems from replacing `key_norm` with DyT, which introduces additional element-wise operations in the recurrent key projection path that are absent in the standard LayerNorm replacement sites of the other architectures.

Table 5: Efficiency comparison between LayerNorm (LN) and Dynamic Tanh (DyT) across architectures grouped by information-flow topology. Throughput measured on a single RTX 5090 with batch size 128 and mixed precision (AMP). Inference memory profiled at batch size 128 in evaluation mode. Relative changes of DyT w.r.t. LN are shown in parentheses.

Topology	Model	Norm	Params (M)	GFLOPs	Throughput (img/s)	Inf. Mem (MB)	Train Mem (MB)
Open-loop	DeiT-T	LN	5.4	0.346	4112.1	109.4	1296.9
		DyT	5.4	0.346	4551.5 (↑10.7%)	109.4	1441.7 (↑11.2%)
	DeiT-S	LN	21.3	1.381	3803.7	272.6	2698.1
		DyT	21.3	1.381	3014.3 (↓20.8%)	272.6	2990.0 (↑10.8%)
Closed-loop	Vim-T	LN	21.6	0.770	2499.6	317.5	2716.3
		DyT	21.6	0.768	2606.5 (↑4.3%)	406.1 (↑27.9%)	2951.9 (↑8.7%)
Hybrid	RWKV-T	LN	5.8	0.373	2023.8	147.8	2369.6
		DyT	5.8	0.369	1816.5 (↓10.2%)	171.1 (↑15.8%)	2924.7 (↑23.4%)

Table 5 reports training-time efficiency alongside inference metrics. Training memory increases consistently across all architectures ($+8.7\%$ to $+23.4\%$), primarily due to caching $\tanh$ intermediate values required for gradient computation during backpropagation. Training throughput exhibits no consistent direction: DeiT-T and Vim-T are slightly faster under DyT ($+10.7\%$ and $+4.3\%$), while DeiT-S and RWKV-T experience slowdowns (-20.8% and -10.2%). The asymmetry likely reflects architecture-specific interactions between the DyT computation graph and GPU kernel scheduling under mixed precision.

These overheads are acceptable at our experimental scale (small models, 32×32 inputs, single GPU) and do not materially affect research iteration speed. However, the consistent memory increase and variable throughput impact should be considered when deploying DyT in larger-scale training pipelines where memory headroom or throughput constraints are binding.

C CIFAR-100-C Corruption Robustness

Table 6: Mean corruption accuracy (%) on CIFAR-100-C [Hendrycks and Dietterich, 2019] at 32×32 resolution (3-seed mean, seeds 42, 123, and 0). All 15 standard corruption types are averaged across severity levels 1–5. Δ reports the difference DyT $-$ LN. Clean Δ reproduces the CIFAR-100 accuracy difference from Table 1 for comparison.

Architecture	Topology	LN Mean Corr. Acc (%)	DyT Mean Corr. Acc (%)	Δ (pp)	Clean Δ (pp)
DeiT-S	open-loop	43.48	47.83	+4.35	+6.90
DeiT-T	open-loop	41.97	46.16	+4.19	+3.91
Vim-T	closed-loop	41.42	48.21	+6.79	+3.45
RWKV-T	hybrid	54.80	57.53	+2.73	−2.17

DyT improves mean corruption accuracy across all four core architectures, with the largest gain for the closed-loop Vim-T ($+6.79$ pp)—nearly double its clean-accuracy improvement ($+3.45$ pp). RWKV-T is the most striking case: despite a -2.17 pp clean-accuracy deficit, DyT still yields a $+2.73$ pp corruption-robustness gain, suggesting an implicit regularization effect that benefits out-of-distribution generalization independently of in-distribution performance.

Table 7 provides the full per-corruption breakdown, grouped by corruption category.

The per-corruption breakdown reveals distinct patterns across topologies. *Noise corruptions* drive the largest gains uniformly: all three architectures improve by $+3.27$ to $+12.51$ pp, with Vim-T and RWKV-T exceeding $+8$ pp on all three noise types. Vim-T is the only architecture where DyT improves accuracy on every single corruption type, consistent with the hypothesis that closed-loop (recurrent) architectures benefit most from the bounded activation range of DyT.

RWKV-T presents a more nuanced picture. While its noise and glass-blur gains are the largest in the table ($+14.61$ pp on glass blur), it degrades on fog (-5.55 pp), contrast (-2.31 pp), pixelate

Table 7: Per-corruption accuracy (%) on CIFAR-100-C (2-seed mean over seeds 42 and 123; severity levels 1–5 averaged per corruption type). Summary table (Table 6) reports 3-seed means. Corruption types are grouped into four categories: Noise, Blur, Weather, and Digital. Δ = DyT $-$ LN; positive values (**bold**) indicate DyT outperforms LN. The final row reports the mean across all 15 corruption types.

	DeiT-S *(open-loop)*			Vim-T *(closed-loop)*			RWKV-T *(hybrid)*		
Corruption	LN	DyT	Δ	LN	DyT	Δ	LN	DyT	Δ
Noise									
Gaussian	30.41	33.77	+**3.36**	25.19	36.82	+**11.63**	35.11	47.62	+**12.51**
Shot	36.41	40.06	+**3.65**	32.07	43.49	+**11.42**	44.15	54.08	+**9.93**
Impulse	31.92	35.19	+**3.27**	24.03	32.79	+**8.76**	32.26	40.53	+**8.27**
Blur									
Defocus	49.53	55.04	+**5.51**	49.51	54.09	+**4.58**	61.68	63.45	+**1.77**
Glass	42.70	42.02	−0.68	31.36	41.31	+**9.95**	37.48	52.09	+**14.61**
Motion	46.88	51.31	+**4.43**	45.14	48.90	+**3.76**	58.19	58.16	−0.03
Zoom	47.02	52.34	+**5.32**	45.41	50.71	+**5.30**	58.49	60.34	+**1.85**
Weather									
Snow	45.30	51.34	+**6.04**	44.16	50.98	+**6.82**	59.87	61.26	+**1.39**
Frost	42.11	48.73	+**6.62**	41.38	48.72	+**7.34**	58.91	60.10	+**1.19**
Fog	45.18	49.93	+**4.75**	46.78	48.52	+**1.74**	63.00	57.45	−5.55
Brightness	50.61	58.56	+**7.95**	53.71	58.58	+**4.87**	69.42	67.46	−1.96
Digital									
Contrast	32.26	38.42	+**6.16**	33.79	36.19	+**2.40**	50.54	48.23	−2.31
Elastic	51.68	56.45	+**4.77**	51.19	55.29	+**4.10**	64.23	64.24	+**0.01**
Pixelate	52.42	53.67	+**1.25**	53.80	56.45	+**2.65**	64.52	62.08	−2.44
JPEG	49.22	52.98	+**3.76**	47.73	55.93	+**8.20**	61.14	64.82	+**3.68**
Mean	43.58	47.99	+**4.41**	41.68	47.92	+**6.24**	54.60	57.46	+**2.86**

(-2.44 pp), and brightness (-1.96 pp). These are corruptions that primarily affect global intensity or low-frequency structure, suggesting that DyT's bounded activation range may clip informative signals in the hybrid architecture's recurrent pathway when the corruption shifts the overall input distribution rather than adding local noise.

Selective DyT corruption robustness (RWKV-T). The selective DyT configuration—preserving `key_norm` as LayerNorm while replacing all other normalization layers—achieves mCA$=55.39\%$ on RWKV-T (2-seed mean, seeds 42 and 123; seed 0 checkpoint unavailable).[1] This falls between LN (54.80%, 3-seed) and full DyT (57.53%, 3-seed), recovering only $+0.59$ pp over LN compared to full DyT's $+2.73$ pp. The result suggests that the corruption-robustness benefit of DyT in RWKV-T is concentrated at the `key_norm` positions: preserving these as LayerNorm eliminates most of the robustness gain, even though full DyT's clean-accuracy deficit (-2.17 pp) is only partially attributable to `key_norm` replacement ($\sim$35%; Table 20). The dissociation between clean accuracy and corruption robustness at the `key_norm` site warrants further investigation.

Table 8: Per-seed mean corruption accuracy (%) for DeiT-S on CIFAR-100-C (15 corruptions $\times$ 5 severities). DyT outperforms LN on every seed, consistent with the clean-accuracy advantage ($\Delta_{\text{clean}} = +6.90$ pp).

Seed	LN mCA (%)	DyT mCA (%)	Δ (pp)
42	43.69	48.57	+4.88
123	43.46	47.40	+3.94
0	43.28	47.51	+4.23
Mean±std	43.48 ± 0.21	47.83 ± 0.65	$+4.35 \pm 0.48$

[1] The seed 0 checkpoint was cleaned after the main experiments and could not be re-evaluated for corruption robustness.

DeiT-S per-seed corruption robustness. DeiT-S—the architecture with the largest clean-accuracy gain (+6.90 pp)—shows a consistent corruption-robustness advantage across all three seeds ($\Delta_{\text{mCA}} = +4.35 \pm 0.48$ pp). The per-seed gains range from $+3.94$ to $+4.88$ pp with no sign reversal, providing strong evidence that DyT's robustness benefit on the open-loop topology is not seed-dependent. The corruption gain is smaller than the clean gain ($+4.35$ vs. $+6.90$ pp), consistent with the pattern noted in Table 6: DyT's regularization effect partially transfers to corrupted inputs, but the benefit attenuates under distribution shift.

Table 9: Per-seed mean corruption accuracy (%) for DeiT-T on CIFAR-100-C (15 corruptions × 5 severities). DyT outperforms LN on every seed, consistent with the clean-accuracy advantage ($\Delta_{\text{clean}} = +3.91$ pp).

Seed	LN mCA (%)	DyT mCA (%)	Δ (pp)
42	41.57	46.56	+4.99
123	42.30	46.38	+4.08
0	42.03	45.53	+3.50
Mean±std	41.97 ± 0.37	46.16 ± 0.55	$+4.19 \pm 0.75$

DeiT-T per-seed corruption robustness. DeiT-T confirms the pattern observed in DeiT-S: DyT improves corruption robustness on every seed ($\Delta_{\text{mCA}} = +4.19 \pm 0.75$ pp), with no sign reversal across seeds. Notably, the corruption gain ($+4.19$ pp) slightly exceeds the clean-accuracy advantage ($+3.91$ pp), suggesting that DyT's robustness benefit is at least as strong as its clean-accuracy benefit on this smaller open-loop architecture.

Table 10: Per-seed mean corruption accuracy (%) for Vim-T on CIFAR-100-C (15 corruptions × 5 severities). DyT outperforms LN on every seed, with the largest per-seed gains among all architectures.

Seed	LN mCA (%)	DyT mCA (%)	Δ (pp)
42	41.20	46.89	+5.69
123	42.16	48.94	+6.78
0	40.90	48.80	+7.90
Mean±std	41.42 ± 0.66	48.21 ± 1.15	$+6.79 \pm 1.11$

Vim-T per-seed corruption robustness. Vim-T shows the largest corruption-robustness advantage among all architectures ($\Delta_{\text{mCA}} = +6.79 \pm 1.11$ pp). The corruption gain substantially exceeds the clean-accuracy advantage ($+6.79$ vs. $+3.45$ pp), indicating that DyT's $\tanh$ compression is particularly effective at stabilizing the bidirectional SSM topology under distribution shift.

Summary: corruption robustness across positive architectures. All three architectures where DyT improves clean accuracy also show consistent corruption-robustness gains, with no sign reversal on any individual seed across the $3 \times 3 = 9$ architecture–seed combinations.

Table 11: Summary of clean and corruption accuracy gains for all positive-Δ architectures. Corruption Δ meets or exceeds clean Δ for DeiT-T and Vim-T, indicating that DyT's robustness advantage is not merely a residual of improved clean performance.

Architecture	Clean Δ (pp)	Corruption Δ (pp)	Ratio
DeiT-S	+6.90	+4.35	0.63
DeiT-T	+3.91	+4.19	1.07
Vim-T	+3.45	+6.79	1.97

Table 11 reveals a striking pattern: while DeiT-S's corruption gain is attenuated relative to its clean gain (ratio 0.63), both DeiT-T and Vim-T exhibit corruption gains that *exceed* their clean-accuracy advantages (ratios 1.07 and 1.97, respectively). This monotonic increase in the corruption-to-clean ratio from open-loop (DeiT) to bidirectional SSM (Vim) suggests that architectures with recurrent state dynamics benefit disproportionately from DyT's bounded activation range under distribution shift, consistent with the hypothesis that $\tanh$ compression prevents error accumulation in recurrent pathways.

Table 12: Per-seed mean corruption accuracy (%) for ConvNeXt-T on CIFAR-100-C (15 corruptions $\times$ 5 severities). LN outperforms DyT on every seed, consistent with the clean-accuracy deficit ($\Delta_{\mathrm{clean}} = -12.11$ pp).

Seed	LN mCA (%)	DyT mCA (%)	Δ (pp)
42	43.66	38.15	−5.51
123	45.38	37.62	−7.76
0	42.55	37.13	−5.42
Mean±std	43.86 ± 1.43	37.63 ± 0.51	-6.23 ± 1.33

ConvNeXt-T per-seed corruption robustness. ConvNeXt-T—the CNN architecture—shows a consistent corruption-robustness deficit across all three seeds ($\Delta_{\mathrm{mCA}} = -6.23 \pm 1.33$ pp), with no sign reversal. The corruption deficit (-6.23 pp) is substantially smaller in magnitude than the clean deficit (-12.11 pp), yielding a corruption-to-clean ratio of 0.51.

Table 13: Per-seed mean corruption accuracy (%) for Swin-T on CIFAR-100-C (15 corruptions $\times$ 5 severities). LN outperforms DyT on every seed, consistent with the clean-accuracy deficit ($\Delta_{\mathrm{clean}} = -17.18$ pp).

Seed	LN mCA (%)	DyT mCA (%)	Δ (pp)
42	44.74	39.58	−5.16
123	44.90	38.70	−6.20
0	44.49	37.53	−6.96
Mean±std	44.71 ± 0.21	38.60 ± 1.03	-6.11 ± 0.90

Swin-T per-seed corruption robustness. Swin-T—the architecture with the largest clean-accuracy deficit (-17.18 pp)—shows a consistent corruption deficit across all seeds ($\Delta_{\mathrm{mCA}} = -6.11 \pm 0.90$ pp). The corruption deficit is dramatically attenuated relative to the clean deficit (ratio 0.36), the lowest among all six architectures, indicating that windowed attention's structural incompatibility with DyT is partially mitigated under corruption.

Table 14: Per-seed mean corruption accuracy (%) for CaiT-XXS24 on CIFAR-100-C (15 corruptions $\times$ 5 severities). LN outperforms DyT on every seed, consistent with the clean-accuracy deficit ($\Delta_{\mathrm{clean}} = -12.34$ pp).

Seed	LN mCA (%)	DyT mCA (%)	Δ (pp)
42	48.00	42.80	−5.20
123	47.92	42.34	−5.58
0	48.25	42.85	−5.40
Mean±std	48.06 ± 0.17	42.66 ± 0.28	-5.40 ± 0.19

CaiT-XXS24 per-seed corruption robustness. CaiT-XXS24—the class-attention architecture—shows the smallest corruption deficit among the three negative architectures ($\Delta_{\mathrm{mCA}} = -5.40 \pm 0.19$ pp), with remarkably low seed variance (± 0.19 pp). The corruption-to-clean ratio of 0.44 follows the same attenuation pattern observed in all negative architectures.

Summary: corruption robustness across all six architectures. All six architectures show directional consistency between clean accuracy and corruption robustness: architectures where DyT improves clean accuracy also show corruption-robustness gains, and architectures where DyT degrades clean accuracy show corruption-robustness deficits. No sign reversal occurs on any of the $6 \times 3 = 18$ architecture–seed combinations.

Table 15: Summary of clean and corruption accuracy changes for all six architectures. The corruption-to-clean ratio is systematically below 1.0 for negative architectures (0.36–0.51), indicating that corruption evaluation attenuates the clean deficit. Positive architectures show ratios from 0.63 to 1.97. Corruption robustness is a modulated echo of the clean-accuracy effect, not an independent signal.

Architecture	Type	Clean Δ (pp)	DyT mCA (%)	LN mCA (%)	Corr. Δ (pp)	Ratio
DeiT-S	Global SA	$+6.90$	47.83 ± 0.65	43.48 ± 0.21	$+4.35$	0.63
DeiT-T	Global SA	$+3.91$	46.16 ± 0.55	41.97 ± 0.37	$+4.19$	1.07
Vim-T	SSM	$+3.45$	48.21 ± 1.15	41.42 ± 0.66	$+6.79$	1.97
CaiT-XXS24	Class Attn	-12.34	42.66 ± 0.28	48.06 ± 0.17	-5.40	0.44
ConvNeXt-T	CNN	-12.11	37.63 ± 0.51	43.86 ± 1.43	-6.23	0.51
Swin-T	Windowed SA	-17.18	38.60 ± 1.03	44.71 ± 0.21	-6.11	0.36

Table 15 reveals that corruption robustness is directionally consistent with clean accuracy across all six architectures: the three positive architectures show corruption gains ($+4.19$ to $+6.79$ pp), and the three negative architectures show corruption deficits (-5.40 to -6.23 pp). Critically, the magnitude of the corruption effect is systematically attenuated relative to the clean effect for all negative architectures (ratios 0.36–0.51), indicating that corruption evaluation produces a modulated echo of the clean-accuracy signal rather than an independent robustness measure. Among positive architectures, the ratio ranges from 0.63 (DeiT-S) to 1.97 (Vim-T), with the monotonic increase from open-loop to recurrent topologies suggesting that $\tanh$ compression provides disproportionate robustness benefits in architectures with recurrent state dynamics.

D ImageNet-100 Resolution Probe

To probe DyT behavior at higher resolution, we conduct an exploratory mechanism probe on ImageNet-100 [Deng et al., 2009] (224×224, 100 classes) using DeiT-S—the architecture where DyT achieves its largest gain at 32×32. These experiments are designed to test whether the capacity limitation analyzed in §K manifests across different α initializations, rather than to establish a competitive baseline.

Setup.

Architecture: DeiT-S (22M parameters, open-loop)

Dataset: ImageNet-100, 224×224, 100 classes

α **initializations:** 0.1, 0.5, 1.0

LN baseline: 70.6% (`in100-deit-s-ln`)

Training: 100 epochs, batch size 64, seed 42, augmented recipe (mixup 0.8, cutmix 1.0, label smoothing 0.1, drop path 0.1)

Results. Table 16 confirms the capacity limitation across three α initializations. Dead-zone duration and final accuracy both improve monotonically with lower α ($22 \to 16 \to 0$ dead-zone epochs; $48.0\% \to 50.2\% \to 52.3\%$ for $\alpha = 1.0, 0.5, 0.1$), but even the best configuration ($\alpha_{\text{init}} = 0.1$, 52.3%) remains -18.3 pp below the LN baseline, confirming insufficient normalization capacity rather than a tuning problem. The mechanism underlying these results is analyzed in §K.

Table 16: ImageNet-100 (224×224) resolution probe: DeiT-S with DyT at varying α_{init} vs. LN baseline. Dead zone denotes the number of initial epochs during which validation accuracy remains near chance ($\sim$1%). This is an exploratory mechanism probe; no tested α value approaches the LN baseline. Lower α yields monotonically better performance, but even the best ($\alpha = 0.1$, 52.3%) remains -18.3 pp below LN.

Configuration	Dead zone (epochs)	Val acc @ep50 (%)	Best val acc (%)	Δ vs LN (pp)
LN baseline	—	—	70.6	—
$\alpha_{\text{init}} = 0.1$	0	39.7	52.3	-18.3
$\alpha_{\text{init}} = 0.5$	16	$\sim$33	50.2	-20.4
$\alpha_{\text{init}} = 1.0$	22	$\sim$1	48.0	-22.6

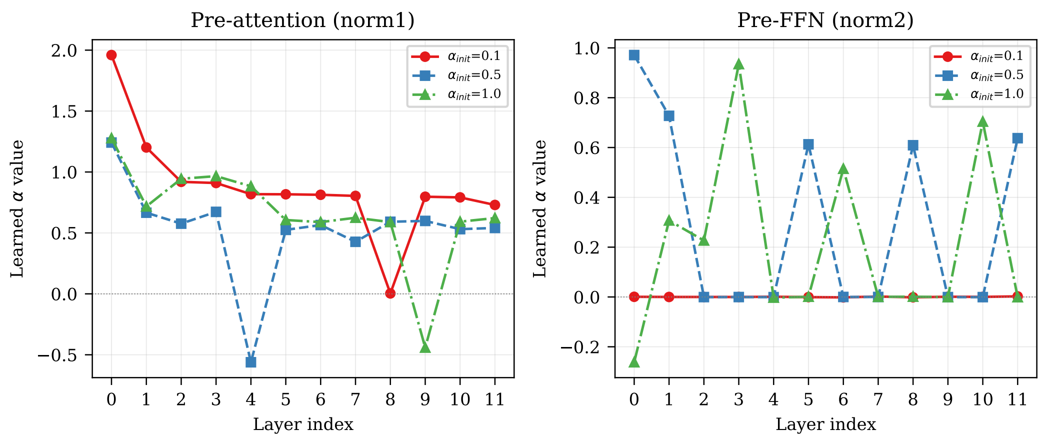

Figure 3: Per-layer learned α values after 100 epochs of training on ImageNet-100 (224×224) for $\alpha_{\text{init}} \in \{0.1, 0.5, 1.0\}$. Left: pre-attention (norm1) α values remain large and positive (0.7–2.0), indicating active scaling. Right: pre-FFN (norm2) α values collapse to near-zero for α_{init}=0.1, while remaining non-negligible for larger initializations, suggesting the FFN path effectively bypasses DyT normalization—a structural asymmetry that may contribute to the capacity limitation observed at 224×224 resolution.

E ConvNeXt-T Probe Experiment

To test the boundary of our topology-based framework, we apply the same universal DyT replacement protocol to ConvNeXt-T [Liu et al., 2022]—a purely convolutional architecture that is topologically open-loop (no recurrent state) but does *not* share the pre-norm residual pattern characteristic of the Transformer-based architectures in our main study. ConvNeXt-T uses LayerNorm positioned post-depthwise-convolution within inverted bottleneck blocks across a hierarchical multi-stage structure, serving feature-scale management rather than pre-norm training stabilization.

Setup. We replace all 23 LayerNorm layers in ConvNeXt-T ($\sim$27.9M parameters) with DyT ($\alpha_{\text{init}} = 0.5$) and train on CIFAR-100 using the same recipe as our main experiments (AdamW, lr $= 1 \times 10^{-3}$, cosine schedule, 100 epochs, batch size 128). All three seeds (42, 123, 0) are run on a single server (westd-32156) with identical codebase to ensure comparability.

Results. DyT replacement produces severe degradation across all three seeds: LN achieves $55.10 \pm 2.45\%$ (seeds: 55.2, 57.5, 52.6%) while DyT achieves $42.99 \pm 0.73\%$ (seeds: 43.66, 43.10, 42.20%), yielding $\Delta = -12.11$ pp. Notably, LN exhibits unusually high inter-seed variance (σ=2.45 pp), substantially above other architectures ($\sigma \leq 1.55$ pp), while DyT variance is low (σ=0.73 pp). This confirms that the topology-dependent pattern (open-loop $\Rightarrow$ DyT beneficial) does not extend to architectures outside the pre-norm residual pattern; the architectural role of normalization, not topology alone, determines DyT compatibility. See §5.1 for discussion of architectural coverage limitations.

F PlainMamba-T: Closed-Loop Counter-Example

To test whether the closed-loop topology universally benefits from DyT, we evaluate PlainMamba-T [Yang et al., 2024]—a simplified Mamba variant without bidirectional scanning—on CIFAR-100 using the same protocol as our main experiments.

Setup. PlainMamba-T ($\sim$12.1M parameters) uses unidirectional selective scan, making it topologically closed-loop like Vim-T. We train with seeds $\{42, 123, 0\}$ using our standard recipe (AdamW, lr $= 1 \times 10^{-3}$, cosine schedule, 100 epochs, batch size 128).

Table 17: PlainMamba-T on CIFAR-100: per-seed and aggregate results. Δ = DyT $-$ LN.

Seed	LN (%)	DyT (%)	Δ (pp)
42	74.13	73.60	-0.53
123	74.10	73.90	-0.20
0	74.30	73.80	-0.50
Mean$\pm$std	74.18 ± 0.11	73.77 ± 0.15	-0.41 ± 0.18

Results. Across three seeds, PlainMamba-T shows a consistent DyT degradation of -0.41 ± 0.18 pp (Table 17). All three seeds yield negative Δ values (-0.53, -0.20, -0.50 pp), confirming that the effect is directionally robust. LN accuracy is remarkably stable ($74.18 \pm 0.11\%$), with a total spread of only 0.20 pp across seeds. Unlike Vim-T ($+3.45$ pp, 3-seed mean), PlainMamba-T shows DyT *degradation* rather than improvement, demonstrating that the closed-loop topology does not uniformly entail DyT benefit.

Interpretation. The discrepancy between Vim-T and PlainMamba-T under DyT may reflect differences in scan pattern (bidirectional vs. unidirectional), internal architecture details, or parameter scale (22M vs. 12.1M). This finding reinforces the N=1-per-topology limitation discussed in §5.1 and motivates treating topology as an organizing lens rather than a deterministic determinant of DyT effectiveness.

G Scale Confound and Controlled Comparison

A potential confound is model scale. Our architectures form two near-parameter-matched pairs that control for this: DeiT-S ($\sim$22M, open-loop) / Vim-T ($\sim$22M, closed-loop) and DeiT-T ($\sim$5.4M, open-loop) / RWKV-T ($\sim$5.8M, hybrid). At the $\sim$22M scale, both architectures benefit from DyT, but the open-loop DeiT-S gains twice as much as the closed-loop Vim-T (the largest gain vs. $+3.45$ pp; Table 1). At the $\sim$5.5M scale, the effect reverses sign entirely: open-loop DeiT-T gains $+3.91$ pp while hybrid RWKV-T loses -2.17 pp. Scale cannot explain a sign reversal at matched capacity, nor a $2\times$ magnitude difference at another matched capacity. Topology is the only factor in our study that consistently covaries with both the direction and magnitude of the DyT effect.

All models are trained from scratch without pretrained weights, producing lower absolute accuracies than transfer-learning baselines. Our DeiT-S result (64.9%) is consistent with prior from-scratch benchmarks: Li et al. [2022] report 65.81% at 200 epochs, and Li et al. [2023] report DeiT-Tiny at 65.08% (200 epochs). Crucially, our study is a *controlled comparison* with normalization type as the only variable; the quantities of interest are the accuracy *differences* (Δ), not absolute values.

H Baseline Accuracy as Confound

One might argue that the decreasing DyT benefit simply reflects a ceiling effect: models with higher baseline accuracy have less room to improve. DeiT-T provides a direct counterexample. Its LN baseline (56.88%) is *lower* than that of DeiT-S (58.0%), yet its DyT gain is also smaller ($+3.91$ pp vs. DeiT-S's larger improvement). Under the ceiling-effect hypothesis, DeiT-T's lower starting accuracy should yield a *larger* Δ, not a smaller one; the observed pattern runs in the opposite direction.

More critically, baseline differences can at most modulate the *magnitude* of improvement; they cannot explain a *sign reversal*. RWKV-T holds the highest baseline among our models (74.63%) and is the only architecture where DyT hurts (-2.17 pp), yet Vim-T still benefits ($+3.45$ pp) despite a substantially higher baseline (61.08%) than DeiT-T (56.88%). If baseline accuracy drove the effect, one would expect a monotonic negative correlation between LN accuracy and Δ; the four architectures exhibit no such relationship.

Recurrent-state topology offers a more parsimonious account: a single categorical variable—the presence and type of recurrent state—correctly accounts for both the direction of the DyT effect across topologies and the sign reversal in the hybrid case in our low-resolution experiments, whereas baseline accuracy fails on both counts.

I Corruption Robustness Discussion

Beyond clean accuracy, DyT improves corruption robustness on CIFAR-100-C (32×32): DyT yields higher mean corruption accuracy than LN across all four architectures (DeiT-S: $+4.35$ pp; DeiT-T: $+4.19$ pp; Vim-T: $+6.79$ pp; RWKV-T: $+2.73$ pp; 3-seed means, Table 6). Notably, RWKV-T benefits from DyT under corruption despite its -2.17 pp clean-accuracy deficit, suggesting an implicit regularization effect under distribution shift. The corruption gain does not simply track clean accuracy: Vim-T's corruption improvement ($+6.79$ pp) nearly doubles its clean gain ($+3.45$ pp), whereas DeiT-S shows the reverse pattern ($+4.35$ pp corruption vs. its larger clean gain), suggesting that the recurrent state in closed-loop architectures may amplify DyT's regularization benefit under shift. Across all model$\times$seed evaluations the direction is consistent, though these results are limited to 32×32 resolution.

J Computational Cost Discussion

Replacing LayerNorm with DyT does not come for free. Table 5 reports training throughput and memory consumption measured on an RTX 5090 (batch size 128, 32×32, AMP). Training memory increases consistently across all architectures (+8.7% to +23.4%), reflecting the additional intermediate activations cached for backpropagation through the $\tanh$ nonlinearity. Training speed, however, shows no consistent direction: DeiT-T and Vim-T are slightly faster with DyT (+10.7% and +4.3%, respectively), while DeiT-S and RWKV-T are slower (-20.8% and -10.2%).

Inference throughput also decreases modestly across all architectures (-2.6% to -15.0%; Table 4), with RWKV-T exhibiting the largest overhead due to DyT replacing the specialized `key_norm` operation. These costs represent a trade-off rather than a free improvement: DyT's accuracy gains in open-loop attention and closed-loop topologies come at a measurable efficiency penalty. In our experimental setting (small models, 32×32 resolution), the overhead is modest and does not materially affect research iteration speed; practitioners considering DyT at larger scales should weigh these costs against accuracy benefits for their specific deployment constraints.

K Exploratory Probe: DyT Capacity at 224×224

The α-sensitivity results in §4.3 reveal topology-dependent fragility on CIFAR-100 (32×32). Our ImageNet-100 experiments at 224×224 expose a more fundamental limitation: **all tested α initializations converge to a narrow accuracy band of approximately 48–52%, falling 18–23 pp below the LN baseline** (70.6%), indicating an apparent capacity limitation of tanh-based normalization substitution that is largely independent of α tuning. The three configurations ($\alpha_{\text{init}} \in \{0.1, 0.5, 1.0\}$) produce final accuracies separated by $\sim$2 pp steps —a negligible spread relative to the 18–23 pp deficit from LayerNorm.

Dead-lock mechanism at high α. Recall that $\mathrm{DyT}(x) = \gamma \cdot \tanh(\alpha x) + \beta$ (Equation 1). The gradients with respect to the input and scale parameter are:

$$\frac{\partial \mathrm{DyT}}{\partial x} = \gamma \cdot \alpha \cdot \mathrm{sech}^2(\alpha x), \tag{2}$$

$$\frac{\partial \mathrm{DyT}}{\partial \alpha} = \gamma \cdot x \cdot \mathrm{sech}^2(\alpha x). \tag{3}$$

When $|\alpha x| \gtrsim 2$, $\mathrm{sech}^2(\alpha x) \to 0$, simultaneously suppressing both gradients. This creates a self-reinforcing dead-lock: saturated activations prevent gradient flow through the layer (Eq. 2), and—critically—α itself cannot update to escape saturation because its own gradient shares the same $\mathrm{sech}^2(\alpha x)$ factor (Eq. 3). The parameter is approximately frozen during the dead zone. Escape occurs indirectly: as other parameters (upstream MLP and attention weights) continue to update, they gradually adjust the pre-activation magnitudes $|x|$, reducing $|\alpha x|$ back into the learnable regime where $\mathrm{sech}^2(\alpha x)$ is non-negligible. Once this happens, α resumes updating and the dead zone ends.

Dead zone as transient effect. Higher α values trigger longer dead zones—22 epochs for $\alpha = 1.0$, 16 for $\alpha = 0.5$, and 0 for $\alpha = 0.1$—during which validation accuracy remains near chance ($\sim$1%). Although dead-zone duration correlates positively with α, this is a transient phenomenon: once training escapes saturation, all configurations converge to the same narrow accuracy band. The $\sim$2 pp separation between adjacent α settings in final accuracy confirms that the initial dead zone does not meaningfully affect asymptotic performance; the dominant bottleneck lies elsewhere.

Bounded output range as the core limitation. Reducing α successfully avoids saturation: $\alpha_{\text{init}} = 0.1$ eliminates the dead zone entirely and achieves the highest DyT accuracy (52.3%). However, this best-case operating point still falls -18.3 pp short of LN (70.6%). The bounded output range of $\tanh$ ($[-1, 1]$) fundamentally limits the normalization capacity available to each layer. LayerNorm computes per-token statistics that adaptively rescale activations across the full feature dimension; DyT's pointwise $\tanh(\alpha x)$ compresses all activations into a fixed interval regardless of their distributional properties. At 224×224 resolution, where pre-activation magnitudes are substantially larger than at 32×32, this capacity gap becomes insurmountable—and no α value can bridge it, as the narrow approximately 48–52% convergence band demonstrates.

Resolution dependence. The onset of dead-lock is controlled by the product $|\alpha \cdot x|$, where x denotes the pre-activation magnitude entering a DyT layer. A patch embedding with patch size p and input channels C_{in} has $\text{fan\_in} = C_{\text{in}} \times p^2$; under Kaiming initialization (`timm` default), the weight standard deviation is set to $1/\sqrt{\text{fan\_in}}$, preserving output variance regardless of p—empirically, the output standard deviation is ≈ 0.33 for all $p \in \{2, 4, 8, 16\}$. Initialization-time embedding magnitudes are therefore not a function of patch size. However, training at 224×224 with $p = 16$ empirically produces larger pre-activation magnitudes than at 32×32 with $p = 4$, entering the saturation regime described above. At 32×32, pre-activations remain in a regime where $\tanh(\alpha x)$ provides effective nonlinear compression without triggering saturation—an effective working interval exists. At 224×224, α values large enough to engage the nonlinear compression of $\tanh$ simultaneously trigger saturation, while values small enough to avoid saturation reduce to the near-linear regime where normalization capacity is insufficient. The result is a resolution-dependent capacity ceiling that cannot be resolved by α tuning alone.

Empirical evidence. We probe this limitation on ImageNet-100 at 224×224 with three α initializations (Table 16). All three converge to a narrow approximately 48–52% band, 18–23 pp below the LN baseline (70.6%, `in100-deit-s-ln`): $\alpha_{\text{init}} = 0.1$ (`in100-deit-s-dyt-a01`) achieves 52.3%

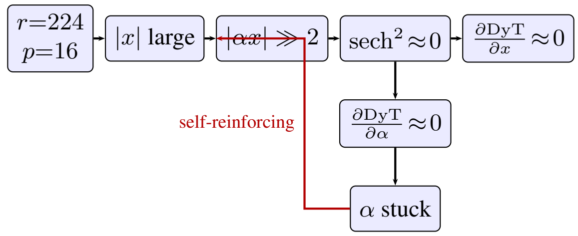

Figure 4: Causal chain of the α dead-lock mechanism at high α. High-resolution inputs produce large pre-activations that saturate $\tanh(\alpha x)$, suppressing gradients for both the signal ($\partial \mathrm{DyT}/\partial x$) and the scale parameter ($\partial \mathrm{DyT}/\partial \alpha$). Because α cannot update, saturation persists—creating a self-reinforcing loop (red arrow). Reducing α breaks this loop but does not recover LN-level performance due to insufficient normalization capacity of the bounded $\tanh$ output (see text).

(-18.3 pp) with no dead zone; $\alpha_{\text{init}} = 0.5$ (`in100-deit-s-dyt-r2`) reaches 50.2% (-20.4 pp) after a 16-epoch dead zone; $\alpha_{\text{init}} = 1.0$ (`in100-deit-s-dyt-a10-r2`) reaches 48.0% (-22.6 pp) after a 22-epoch dead zone. The uniform $\sim$2 pp spacing between adjacent α settings, combined with the consistently large deficit from LN, indicates that α selection modulates performance only marginally within a regime bounded far below LN capacity. By contrast, $\alpha_{\text{init}} = 0.5$ on CIFAR-100 (32×32) yields 64.9% (3-seed mean)—a $+6.90$ pp gain over LN (58.0%)—confirming that the limitation is resolution-specific, not inherent to any particular α value.

Scope and interpretation. This capacity limitation is observed in our ImageNet-100 experiment with a single architecture (DeiT-S), one higher-resolution dataset, and three α values; we present it as an exploratory mechanistic probe of DyT's resolution boundary rather than a general claim about tanh-based normalization at scale. Our primary conclusions regarding topology-dependent DyT effectiveness are validated in the 32–64 px regime where DyT operates within its effective capacity; extending these findings to higher resolutions remains an open challenge. Whether activation functions with unbounded or adaptive-range outputs can close the gap at 224×224 and beyond is an important direction for future work.

L Small-Scale Experiments as Controlled Testbeds

Our primary experiments operate at 32×32 (CIFAR-100) and 64×64 (Tiny-ImageNet)—substantially below the 224×224 standard in modern vision benchmarks. This design is deliberate: small-scale controlled settings isolate normalization type as the sole variable while minimizing confounders from dataset scale, training cost, and hyperparameter sensitivity. The resulting computational efficiency enables our multi-seed, multi-architecture factorial protocol; replicating the same design at ImageNet scale would require approximately $50\times$ the compute budget, precluding the systematic comparison across four architectures and three seeds that underpins our topology analysis. This experimental strategy has extensive precedent: the original ViT paper [Dosovitskiy et al., 2021] includes CIFAR-10/100 experiments alongside ImageNet results, and studies of training dynamics, normalization effects, and architectural design principles routinely employ small-scale benchmarks to establish controlled findings before scaling.

The mechanistic understanding derived from these experiments extends beyond the specific resolution. Our topology-dependent spectrum—DyT beneficial in open-loop attention architectures, diminishing in closed-loop, harmful in hybrid—identifies recurrent state coupling as a design axis for normalization-free architectures, a structural insight independent of input resolution. Equally important, our ImageNet-100 probe (§K) demonstrates that DyT effectiveness is *not* trivially transferable across resolutions: the α dead-lock and insufficient normalization capacity at 224×224 reveal failure modes that small-scale experiments alone cannot predict. This negative result carries direct practical value—it prevents practitioners from naïvely extrapolating small-scale DyT gains to higher-resolution settings without accounting for the resolution-dependent capacity limitations identified in §K.

M Per-Seed Tiny-ImageNet Results

Table 18 reports individual seed results. Vim-T shows the largest per-seed variance (Δ spans -1.19 to $+4.40$ pp), with one sign reversal, aligning with its narrow α-stability window (§4.3). DeiT-S and RWKV-T show consistent sign across all seeds. The closed-loop Tiny-ImageNet result should be characterized as marginal rather than robust.

Table 18: Per-seed Tiny-ImageNet accuracy (%) for DyT vs. LayerNorm. Δ = DyT $-$ LN. All architectures include 3 seeds each (42, 123, 0). Per-seed accuracy values are rounded to one decimal place; mean Δ is computed from unrounded values.

Architecture	**Seed**	**DyT**	**LN**	**Δ (pp)**
DeiT-S (open)	42	46.4	40.6	$+5.80$
	123	46.1	39.2	$+6.90$
	0	45.9	39.6	$+6.34$
	Mean±std	46.2 ± 0.3	39.8 ± 0.7	$+6.35$
DeiT-T (open)	42	41.8	41.2	$+0.59$
	123	42.0	41.7	$+0.36$
	0	42.5	41.3	$+1.12$
	Mean±std	42.1 ± 0.3	41.4 ± 0.3	$+0.69$
Vim-T (closed)	42	46.9	42.5	$+4.40$
	123	44.2	45.4	-1.19
	0	42.5	41.5	$+1.00$
	Mean±std	44.5 ± 2.2	43.1 ± 2.0	$+1.40$
RWKV-T (hybrid)	42	53.4	54.9	-1.50
	123	51.6	54.7	-3.10
	0	52.7	54.3	-1.55
	Mean±std	52.6 ± 0.9	54.6 ± 0.3	-2.05

N Sensitivity to Initialization Scale (α)

O Training Collapse in Closed-Loop DyT

DyT-induced training failures partition into three modes: (1) *learn-then-collapse*—RMT-T (single seed, single configuration) reaches $\sim$30% accuracy before collapsing to $\sim$2% at epoch 16; (2) *never-learned*—Sequencer2D-S (single seed, single configuration) never exceeds $\sim$5% under DyT ($\Delta = -55.9$ pp); (3) *seed-dependent*—Vim-T on Tiny-ImageNet collapses in 1 out of all 5 tested seeds at epoch 3, entering a deadlock phase (loss $\approx \ln(200)$) before recovering at epoch 50. These modes differ in onset, reversibility, and determinism, consistent with potentially distinct instability mechanisms, though further seeds and configurations are needed to confirm.

The Vim-T collapse affects 1 out of all 5 tested seeds (20% incidence); the $+1.40$ pp mean in Table 23 uses the standard 3-seed set (seeds 42, 123, 0). Including all five seeds yields -2.19 pp, underscoring the outsized impact of a single catastrophic initialization. This constitutes a tail risk of the closed-loop + DyT combination rather than systemic failure. Per-seed breakdown (Tiny-ImageNet, best validation accuracy): seed 42: LN 42.49%, DyT 46.88% ($+4.39$); seed 123: LN 45.43%, DyT 44.21% (-1.22); seed 0: LN 41.54%, DyT 42.51% ($+0.97$); seed 3: LN 41.20%, DyT 21.37% (-19.83, collapsed); seed 4: LN 39.81%, DyT 44.46% ($+4.65$). PlainMamba-Wide + DyT shows a distinct pattern: deterministic collapse (3/3 seeds), shifted to stochastic by adding MLP (§5).

P Gradient Saturation Analysis

We examine gradient-level mechanisms using 1000-sample gradient flow measurements from converged checkpoints (seed = 42).

Table 19: α-sensitivity sweep on CIFAR-100. Vim-T (closed-loop) collapses at 4 of 5 tested α values; DeiT-S (open-loop) converges at $\alpha \leq 0.50$, degrades at α=1.0, and collapses at α=2.0; RWKV-T (hybrid) converges at all tested values. RWKV-T performance improves with higher α, largely closing the DyT–LN gap; Swin-T (windowed attention) peaks near α=1.0 but remains 9.8 pp below LN; CaiT-XXS24 (deep self-attention) improves gradually with higher α (45.98–51.33%, basic recipe) but remains $\geq$6.57 pp below the basic-recipe LN baseline across all four tested values. A run is classified as "collapsed" if its best validation accuracy remains at or below chance level ($\leq 3\%$ for CIFAR-100's 100 classes); "degraded" denotes accuracy well above chance but substantially below the best converged result for that architecture. §3-seed mean±std (seeds 42, 0, 123); all other entries are single seed (seed = 42). †Single-seed probe (see Table 1). ‡Non-standard hyperparameters.

Architecture	α	**Val Acc (%)**	**Outcome**	**Exp. ID**
Vim-T (closed-loop)	0.25	$\sim$1	Collapsed	`alpha-vim-a025`
	0.50	63.7	Converged	`alpha-vim-a050`
	0.75	$\sim$2	Collapsed	`alpha-vim-a075`
	1.00	$\sim$3	Collapsed	`alpha-vim-a100`
	2.00	$\sim$2	Collapsed	`alpha-vim-a200`
DeiT-S (open-loop)	0.10	61.8	Converged	`alpha-deit-a010`
	0.25	63.6	Converged	`alpha-deit-a025`
	0.50	65.1	Converged	`alpha-deit-a050`
	1.00	$\sim$30	Degraded	`alpha-deit-a100`
	2.00	$\sim$1	Collapsed	`alpha-deit-a200`
RWKV-T (hybrid)	0.10	61.8	Converged	`alpha-rwkv-a010`
	0.25	68.7	Converged	`alpha-rwkv-a025`
	0.50	72.26	Converged	`pvtB-1b`
	1.00	$74.42 \pm 0.14^{\S}$	Converged	`alpha-rwkv-a100`
	2.00	74.09	Converged	`alpha-rwkv-a200`
Swin-T†‡ (windowed attn)	0.30	40.61	Converged	`swin-t-dyt-a03-s42`
	0.50	46.73	Converged	`swin-t-dyt-s42-r3`
	1.00	54.26	Converged	`swin-t-dyt-a10-s42`
	2.00	53.31	Converged	`swin-t-dyt-a20-s42`
CaiT-XXS24† (deep attn)	0.30	45.98	Converged	`cait-xxs24-dyt-a03-s42`
	0.50	49.5	Converged	`cait-xxs24-dyt-s42`
	1.00	50.7	Converged	`cait-xxs24-dyt-a10-s42`
	2.00	51.33	Converged	`cait-xxs24-dyt-a20-s42`

Depth-dependent α growth. Learned α increases with depth and inverse width: DeiT-S (12L/d=384) maintains $\alpha = 0.27$; DeiT-T (12L/d=192) learns $\alpha = 0.47$; CaiT-XXS24 (24+2L/d=192) reaches $\alpha = 0.60$–0.85 in deep layers, with the final norm at $\alpha = 6.35$.

The saturation mechanism. High α drives saturation where $\text{sech}^2(\alpha x) \to 0$ suppresses gradients: CaiT reaches 68.3% saturation, vs. 12.4% for DeiT-S, 10.5% for DeiT-T, and 3.9% for Vim-T. Within CaiT, DyT exhibits $3.21\times$ gradient decay across SA blocks vs. $1.79\times$ under LN ($1.8\times$ steeper), with per-layer DyT/LN ratios declining from 0.49 (SA-0) to 0.27 (SA-23).

Width buffer. DeiT-T learns $\alpha = 0.468$ vs. DeiT-S $\alpha = 0.269$ at matched depth—a $1.74\times$ ratio, providing a gradient-level account for the width effect: wider layers maintain lower α and saturation.

Open question: Vim-T. Vim-T shows the lowest saturation (3.9%) yet steep gradient decay ($8.02\times$ under DyT vs. $3.29\times$ under LN), while still gaining $+3.45$ pp. The mechanism remains unclear; bidirectional SSM dynamics and MLP compensation are candidate explanations.

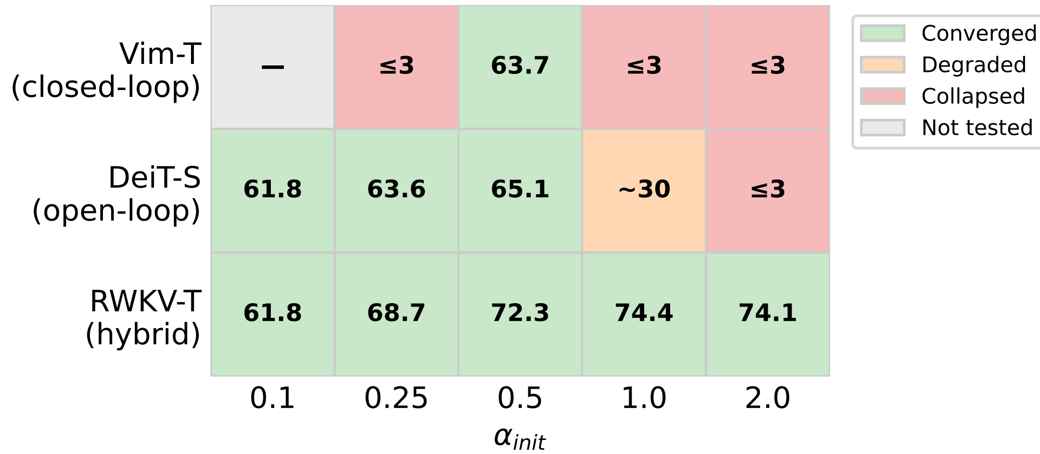

Figure 5: α-sensitivity heatmap across topologies on CIFAR-100. Green cells indicate successful convergence (with accuracy shown); red cells indicate training collapse ($\leq 3\%$ accuracy); gray cells were not tested. Each topology exhibits a distinct α-stability window: RWKV-T (hybrid) converges at all tested values; DeiT-S (open-loop) converges at $\alpha \leq 0.50$, degrades at α=1.0, and collapses at α=2.0; Vim-T (closed-loop) converges only at α=0.50; Swin-T (windowed attention) converges at all four tested values with performance peaking near α=1.0; CaiT-XXS24 (deep self-attention) converges at all four tested values with gradual improvement at higher α (basic recipe), but remains $\geq$6.57 pp below the basic-recipe LN baseline. Single seed (seed = 42) except where noted.

Q RWKV-T Decomposition Analysis

Given that RWKV-T is the only multi-seed core architecture where DyT degrades performance, we perform a targeted ablation to identify which normalization layers contribute to this degradation. Specifically, we compare three configurations: (1) the full LayerNorm baseline, (2) DyT replacement of all norms *except* `key_norm`, and (3) full DyT replacement including `key_norm`. Results are reported across three seeds (42, 123, 0).

Table 20: Decomposition of the DyT performance gap in RWKV-T on CIFAR-100. `DyT (excl. key_norm)` replaces all norms except `key_norm` with DyT. Structural norms account for $\sim$65% and `key_norm` for $\sim$35% of the total degradation (3-seed mean), though the relative contribution varies across seeds (structural: 48–83%).

Configuration	Seed	Acc (%)	Δ vs Full LN (pp)	Gap Explained
Full LN (baseline)	42	74.46	—	—
DyT (excl. key_norm)	42	73.13	-1.33	Structural: 60%
Full DyT	42	72.26	-2.20	+ key_norm: 40%
Full LN (baseline)	123	74.52	—	—
DyT (excl. key_norm)	123	73.54	-0.98	Structural: 48%
Full DyT	123	72.50	-2.02	+ key_norm: 52%
Full LN (baseline)	0	74.90	—	—
DyT (excl. key_norm)	0	73.00	-1.90	Structural: 83%
Full DyT	0	72.60	-2.30	+ key_norm: 17%
3-seed mean±std			-2.17 ± 0.14	Structural: 64 ± 17% / key_norm: 36 ± 17%

Table 20 shows the 2.17 ± 0.14 pp degradation decomposes into two components: preserving `key_norm` recovers 0.77 ± 0.33 pp ($\sim$35%), while structural norms account for 1.41 ± 0.46 pp ($\sim$65%). The `key_norm` contribution varies across seeds (42: 40%; 123: 52%; 0: 17%), but structural norms dominate in two of three.

This reveals two distinct incompatibility sources: `key_norm` enforces a magnitude invariant that DyT cannot replicate, while the broader structural gap indicates DyT is incompatible with the hybrid architecture beyond the specialized `key_norm` site. Selective replacement preserving architecture-specific functional norms may mitigate degradation.

R MLP Compensation Ablation

The MLP ablation reveals two findings that together identify MLP as the critical enabler of DyT in recurrent architectures. First, removing MLP sub-layers from Vim-T converts DyT's $+3.45$ pp advantage into a -8.09 pp deficit—a complete Δ reversal ($\sim$12 pp swing)—exposing MLP as a *functional compensation layer* that absorbs the feature distribution management DyT cannot perform. Second, this reversal is accompanied by a $6.3\times$ inflation in inter-seed variance (σ=6.66 pp vs. 1.06 pp for LN), identifying MLP as DyT's *stability anchor*. We validate via bidirectional ablations on Vim-T (natively includes MLP) and PlainMamba-T (does not). Most conditions report 3-seed means; the PlainMamba-T +MLP ablation is single-seed (noted in Table 21).

Table 21: MLP compensation ablation on CIFAR-100 (3-seed means $\pm$ std where available). In the full Vim-T model, DyT outperforms LN by $+3.45$ pp. Removing MLP *reverses* this gap to -8.09 pp—a 12 pp swing—while inflating DyT inter-seed variance to σ=6.66 pp (6.3$\times$ that of LN). Adding MLP to PlainMamba-T does not confer DyT tolerance.

Architecture	Variant	DyT (%)	LN (%)	Δ (pp)
Vim-T	Full (with MLP)	64.53 ± 0.81	61.08 ± 0.60	$+3.45$
Vim-T	$-$MLP	50.83 ± 6.66	58.92 ± 1.06	-8.09
PlainMamba-T	Original (no MLP)	73.60	74.13	-0.53
PlainMamba-T	+MLP	63.40	64.50	-1.10

All no-MLP conditions report 3-seed means (s42, s0, s123).

Δ reversal: MLP as functional compensation. In the full Vim-T model, DyT outperforms LN by $+3.45$ pp (64.53% vs. 61.08%).Removing MLP sub-layers *reverses* this advantage to $\Delta = -8.09$ pp (50.83 ± 6.66% vs. 58.92 ± 1.06%).The total swing from $+3.45$ to -8.09 pp ($\sim$12 pp) is recipe-independent: both conditions use identical training recipes, ruling out augmentation confounds. The asymmetry is stark: LN degrades modestly without MLP (61.08% $\rightarrow$ 58.92%, -2.16 pp), while DyT collapses (64.53% $\rightarrow$ 50.83%, -13.70 pp). This identifies MLP as a *functional compensation layer*—it provides the activation distribution management that LayerNorm handles intrinsically but that DyT's $\tanh(\alpha x)$ scaling cannot.

Instability amplification: MLP as stability anchor. The Δ reversal is accompanied by a qualitative change in training stability. DyT no-MLP exhibits σ=6.66 pp across three seeds (43.92/51.37/57.20%),more than $6.3\times$ the LN no-MLP variance (σ=1.06 pp).This exceeds the highest 3-seed variance in Table 1 (σ=2.5 pp for ConvNeXt-T LN) by $2.7\times$. The three seeds produce qualitatively distinct outcomes—early collapse (s42: 43.92%), severe overfitting (s0: 51.37%), and stable convergence (s123: 57.20%)—indicating that without MLP, seed-dependent initialization determines the training regime entirely. LN maintains σ=1.06 pp without MLP, confirming that the instability is specific to normalization-free operation rather than an intrinsic consequence of MLP removal.

Control: MLP is necessary but not sufficient. Grafting MLP onto PlainMamba-T does not confer DyT tolerance ($\Delta = -1.10$ pp),implicating Vim-T's bidirectional scan as an independent factor (§5). The PlainMamba-T +MLP result is a single-seed exploratory finding and should be interpreted with appropriate caution.

Dual role of MLP. The Δ reversal and instability amplification are dual manifestations of MLP's role for normalization-free operation. DyT's $\tanh(\alpha x)$ scaling, lacking explicit mean-variance normalization, depends on MLP sub-layers to absorb activation distribution shifts that accumulate through recurrent state transitions. Without this compensation: (1) performance inverts (Δ: $+3.45 \rightarrow -8.09$ pp), because the distribution management that LN handles intrinsically goes unaddressed; and (2) the optimization landscape fragments (σ=6.66 pp, 6.3× LN), because initialization encounters qualitatively different loss surface regions. LN requires neither role—it maintains both performance (58.92% vs. 61.08% full, −2.16 pp) and stability (σ=1.06 pp) without MLP.

S Three-Factor Analysis Summary

Table 22 extends the three-factor analysis from §5 (Table 2) to all ten architectures. The empirical pattern is consistent with six of ten cases. ConvNeXt-T's depthwise-separable structure produces extreme rank collapse (−51.4%) despite moderate α (0.42), indicating geometric incompatibility outside the α–saturation pathway. RWKV-T decomposes into structural (∼65%) and `key_norm` (∼35%) components (Table 20).

Table 22: Extended three-factor analysis (all ten architectures). †Single seed. ‡Non-standard hyperparameters (see Table 1). *ConvNeXt's 1×1 pointwise convolution is functionally analogous to MLP but operates within a depthwise-separable structure incompatible with DyT. §ConvNeXt-T LN exhibits high inter-seed variance (σ=2.45 pp).

Architecture	Topology	Depth / Width	MLP	Δ (pp)	Dominant Factor
DeiT-S	Open (attn)	12 / 384	✓	+6.90	Shallow-wide buffer
DeiT-T	Open (attn)	12 / 192	✓	+3.91	Narrow width
CaiT-XXS24	Open (attn)	26 / 192	✓	−12.34	Deep-narrow compression
ConvNeXt-T§	Open (CNN)	—	✓*	−12.11	Structural mismatch
Swin-T	Open (windowed attn)	4-stage / —	✓	−17.18‡	Resolution confound
Vim-T	Closed (SSM)	10 / 384	✓	+3.45	SSM + MLP compensation
PlainMamba-T	Closed (SSM)	12 / 384	—	−0.41	No MLP channel
RWKV-T	Hybrid	12 / —	✓	−2.17	Structural + `key_norm`
RMT-T	Open (retention)	14 / 64-512	✓	collapse†	Retention mech. collapse
Seq2D-S	Closed (BiLSTM)	21 / 192	—	collapse†	BiLSTM recurrence

Topology shapes how tanh compression propagates. In open-loop attention, compression affects per-layer inputs without cross-layer state accumulation; shallow-wide architectures absorb this. In closed-loop SSMs, recursive state propagation carries compression across positions, making MLP compensation critical. The hybrid RWKV-T introduces a third mode via `key_norm`-specific vulnerability. The α-stability window is itself topology-dependent (§4.3).

T Cross-Dataset Validation

To assess whether the topology-dependent pattern observed on CIFAR-100 generalizes beyond a single dataset, we evaluate the core four multi-seed architectures on Tiny-ImageNet (200 classes, 64×64 images resized to 32×32, 100K training images).

Table 23 confirms the topology-dependent pattern on Tiny-ImageNet: DeiT-S maintains +6.35 pp, DeiT-T shows attenuated but direction-consistent +0.69 pp (3 seeds), Vim-T is marginal (+1.40 pp, with seed-dependent sign reversal[2]; Appendix M), and RWKV-T degrades consistently (−2.05 pp). Both datasets use 32 × 32 inputs; whether the pattern persists at higher resolutions remains open.

Per-seed results are reported in Appendix M.

Extended probes: native resolution and windowed attention. Two single-seed probes extend the cross-dataset analysis to conditions not covered by the multi-seed experiments above: DeiT-S

[2] Vim-T mean is +1.40 pp, but seed 123 shows −1.19 pp; remaining seeds: +4.40 and +1.00 pp.

Table 23: Cross-dataset validation on Tiny-ImageNet. All architectures averaged over 3 seeds (42, 123, 0). CIFAR-100 Δ shown for comparison. †Per-seed signs disagree (seed 42: $+4.40$ pp, seed 123: -1.19 pp, seed 0: $+1.00$ pp).

Architecture	LN Acc (%)	DyT Acc (%)	Δ (pp)	CIFAR-100 Δ
DeiT-S	39.80 ± 0.72	46.15 ± 0.26	$+\mathbf{6.35}$	$+6.90$
DeiT-T	41.40 ± 0.26	42.09 ± 0.34	$+0.69$	$+3.91$
Vim-T	43.13 ± 2.03	44.54 ± 2.22	$+1.40^\dagger$	$+3.45$
RWKV-T	54.62 ± 0.33	52.57 ± 0.91	-2.05	-2.17

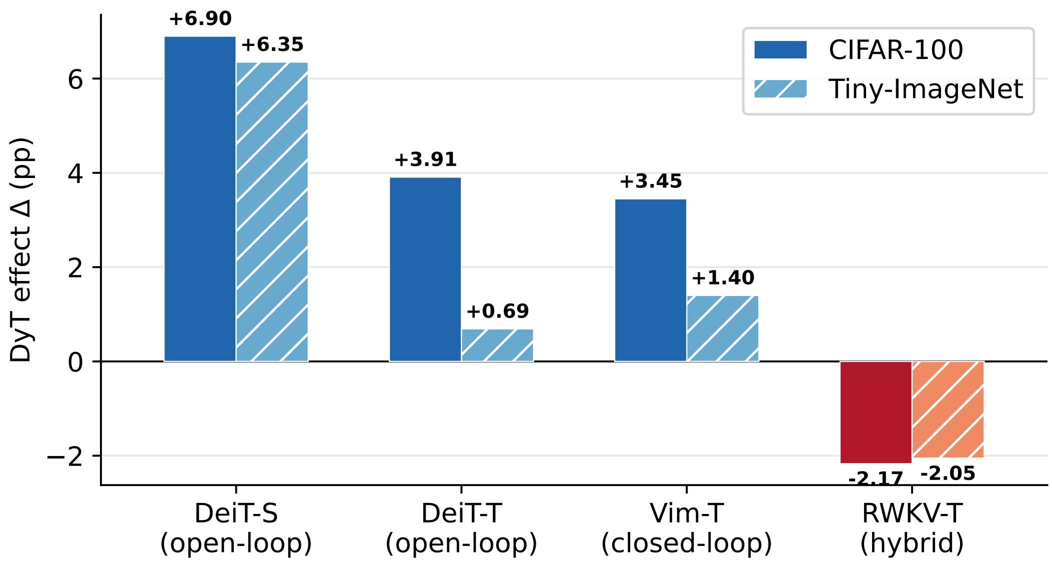

Figure 6: DyT effect (Δ accuracy) on CIFAR-100 vs. Tiny-ImageNet. The topology-dependent pattern—global standard self-attention benefits, hybrid hurts—replicates across both datasets. DeiT-T's Tiny-ImageNet gain ($+0.69$ pp, 3 seeds) is substantially attenuated from its CIFAR-100 result ($+3.91$ pp).

at Tiny-ImageNet's native 64×64 resolution ($4\times$ the spatial area of the 32×32 runs) and Swin-T at 32×32.

At 64×64, DeiT-S with DyT achieves 53.93% vs. 52.46% for LN ($\Delta = +1.47$ pp, single seed), directionally consistent with the CIFAR-100 result ($+6.90$ pp) but substantially attenuated. The attenuation coincides with a striking regularization asymmetry: DyT reduces the train–validation gap from 25.21 pp (LN) to 7.74 pp—a $3.3\times$ compression. This pattern suggests that DyT provides strong implicit regularization that suppresses overfitting, but may simultaneously constrain capacity exploitation at higher resolution, producing the narrower accuracy margin. On this reading, the effect attenuation is not evidence against the topology-dependent pattern but rather a consequence of DyT's dual role as both a normalization substitute and an implicit regularizer whose strength becomes resolution-dependent.

Swin-T on 32×32 Tiny-ImageNet shows $\Delta = -9.6$ pp (33.0% DyT vs. 42.6% LN), directionally consistent with CIFAR-100 (-17.18 pp). DyT training accuracy reaches only 53.5% vs. LN's 87.1%, indicating the same systematic underfitting observed on CIFAR-100 persists across datasets for windowed self-attention.

Both probes are single-seed and serve as directional evidence rather than quantitative estimates of effect size. The sign consistency across two datasets—positive Δ for global self-attention, negative for windowed self-attention—reinforces the topology-dependent pattern established in the multi-seed analysis.

U RMSNorm Control Analysis

To distinguish topology-intrinsic normalization dependence from DyT-specific artifacts, we introduce RMSNorm as a control condition on two architectures representing opposite ends of the topology spectrum: DeiT-S (open-loop, global SA) and RWKV-T (hybrid). RMSNorm removes mean-centering while retaining variance normalization, providing an intermediate point between full LayerNorm and DyT's complete normalization removal.

Open-loop (DeiT-S): warmup-dependent benefit from normalization removal. The three conditions form a strict ordering: DyT (64.9%) $>$ RMSNorm ($60.7 \pm 0.3\%^{w}$) $>$ LN (58.0%). Removing mean-centering alone (LN $\rightarrow$ RMSNorm) yields $+2.7$ pp; further removing variance normalization and introducing learned $\tanh$ gating (RMSNorm $\rightarrow$ DyT) yields an additional $+4.2$ pp. With warmup, each step of normalization removal improves performance; however, this ordering is warmup-dependent (see below). (The RMSNorm runs used 5-epoch warmup and are marked with superscript ^{w}. A matched-warmup control with warmup$=$0 across 3 seeds yields $57.8 \pm 1.9\%$ (seeds 42/0/123: 59.5/55.7/58.3%) vs. $60.7 \pm 0.3\%$ with warmup$=$5—a 2.9 pp mean deficit. Notably, removing warmup amplifies cross-seed standard deviation by $\sim$6$\times$ (from 0.3 pp to 1.9 pp), indicating that warmup scheduling plays a significant role in stabilizing training for normalization-free alternatives. With warmup$=$0, the RMSNorm mean (57.8%) falls at the LN baseline (58.0%), so the ordering LN $<$ RMSNorm $<$ DyT is maintained only when warmup is included.)

Hybrid (RWKV-T): uniform degradation from any normalization removal. Both alternatives degrade comparably: LN ($74.6 \pm 0.2\%$) $\gg$ RMSNorm ($72.4 \pm 0.2\%$) $\approx$ DyT ($72.5 \pm 0.2\%$), all 3 seeds. The RMSNorm–DyT difference is 0.04 pp—within noise—while both fall $\sim$2 pp below LN. The degradation is triggered by the loss of mean-centering (the operation common to both RMSNorm and DyT relative to LN), not by DyT's $\tanh$ gating mechanism. DyT's learned α and β parameters confer neither additional benefit nor additional harm beyond the initial normalization removal.

Core finding: degradation is topology-determined, not method-specific. The contrast establishes that the pattern in Table 1 is *normalization-intrinsic*: in DeiT-S, normalization removal of any degree improves performance; in RWKV-T, normalization removal of any degree degrades performance. The replacement method (RMSNorm vs. DyT) modulates magnitude but not direction. This rules out explanations based on DyT's parameterization (e.g., $\tanh$ saturation, α initialization sensitivity) and implicates architectural information flow as the determining factor—specifically, whether the topology permits global context aggregation that renders explicit statistical normalization redundant.

V Mechanism Integration: α–Saturation–Rank Correlation

This correlational analysis uses converged checkpoints from four architectures (seed$=$42). Two pathways emerge (Figures 8, 7): (i) low learned α (DeiT-S: 0.25, Vim-T: 0.29) keeps tanh near-linear, preserving rank (-3.8% and $+12.9\%$); (ii) high α (RWKV-T: 1.17) drives saturation, compressing rank by -13.8%. Vim-T uniquely *increases* rank under DyT, suggesting its selective scan redistributes rather than loses information. Rank compression is not beneficial regularization: DeiT-T shows larger compression (-27.7%) than DeiT-S (-3.8%) yet smaller accuracy gain ($+3.91$ vs. $+6.90$ pp).

ConvNeXt-T illustrates a distinct failure: despite moderate α (0.42), its depthwise-separable convolution produces -51.4% rank collapse, indicating structural mismatch outside the α–saturation pathway.

Gradient saturation. Gradient-level analysis (Appendix P) confirms the compression mechanism: learned α increases with depth and inverse width, driving saturation fractions from 3.9% (Vim-T) to 68.3% (CaiT-XXS24), consistent with the depth–width account above.

DyT-induced training failures partition into three modes (learn-then-collapse, never-learned, seed-dependent); full analysis is in Appendix O.

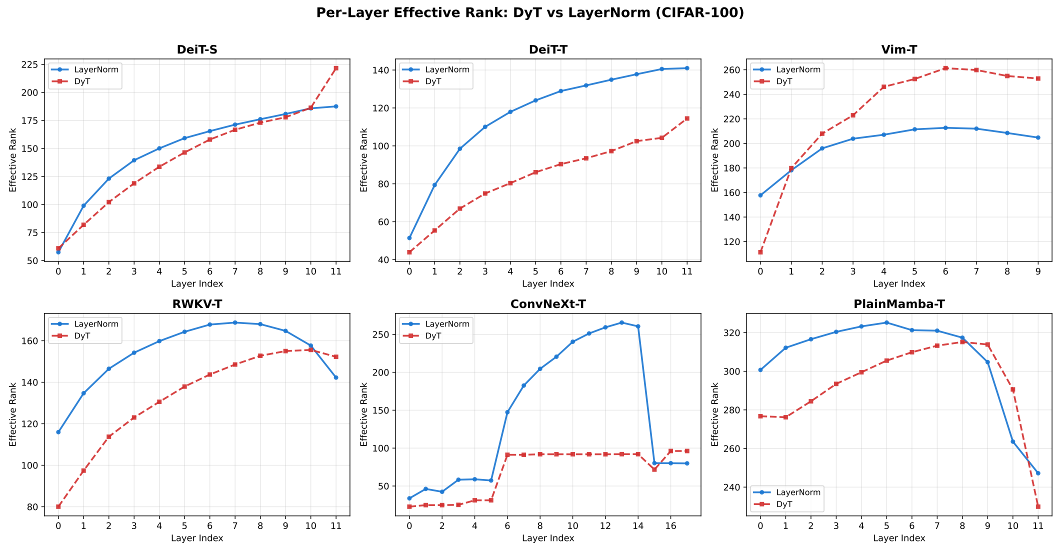

Figure 7: Effective rank change (DyT vs. LN) across architectures. Negative values indicate representational compression under DyT.

W Detailed Mechanism Analyses

W.1 Scan Direction and Information Aggregation

Beyond MLP availability, Vim-T and PlainMamba-T differ fundamentally in how they aggregate information across the input sequence. Vim-T employs bidirectional Mamba scanning [Zhu et al., 2024], where each token simultaneously receives forward and backward context—producing representations that aggregate information from the entire input sequence, achieving a form of global context integration distinct from—but comparable in scope to—self-attention. PlainMamba-T uses unidirectional scanning with learned directional biases [Yang et al., 2024], restricting each position to a single-direction context. This architectural distinction maps onto the broader pattern in Table 1: architectures where every token aggregates information globally (DeiT-S/T, Vim-T) show positive DyT improvements on CIFAR-100 (direction-consistent across all seeds), while those relying on local or sequential processing degrade—though to varying degrees: PlainMamba-T (-0.41 pp) is practically neutral, RWKV-T (-2.17 pp) shows mild degradation, and Swin-T (-17.18 pp) and ConvNeXt-T (-12.11 pp) suffer catastrophic failure. Notably, RWKV-T and PlainMamba-T are both unidirectional sequential architectures, yet differ by $5.3\times$ in Δ. Decomposing RWKV-T's deficit via `key_norm` ablation (Table 20) into structural ($\sim65\%$, ≈-1.41 pp) and `key_norm`-specific ($\sim35\%$) components brings the structural component closer to PlainMamba-T's -0.41 pp; the residual gap is attributable to differences in token-mixing—RWKV-T's WKV attention performs data-dependent weighted averaging, whereas PlainMamba-T's SSM recurrence applies fixed-structure state transitions. The magnitude gap between partial-sequence architectures—catastrophic for windowed/local models (Swin-T, ConvNeXt-T) vs. mild for sequential SSMs—is further explained by α dead-lock (§K): windowed and local architectures not only lack full-sequence statistics but also trigger tanh saturation from high-resolution patch embeddings, producing catastrophic failure rather than gradual degradation.

We hypothesize that DyT's learnable $\tanh(\alpha x)$ can substitute for LN's statistical normalization when the preceding layer produces globally-aggregated representations: the full-sequence statistics that LN would compute are implicitly available in the activations. Under unidirectional or windowed processing, activations reflect partial-sequence statistics, making LN's explicit normalization non-redundant. We note that this hypothesis addresses LN's role in aggregating cross-token statistics; regarding LN's separate contribution to gradient normalization, a functional substitution relationship exists with DyT's $\tanh$-induced implicit gradient clipping: in open-loop architectures, this clipping suffices to replace LN's gradient normalization, consistent with DeiT-S/T showing positive Δ under

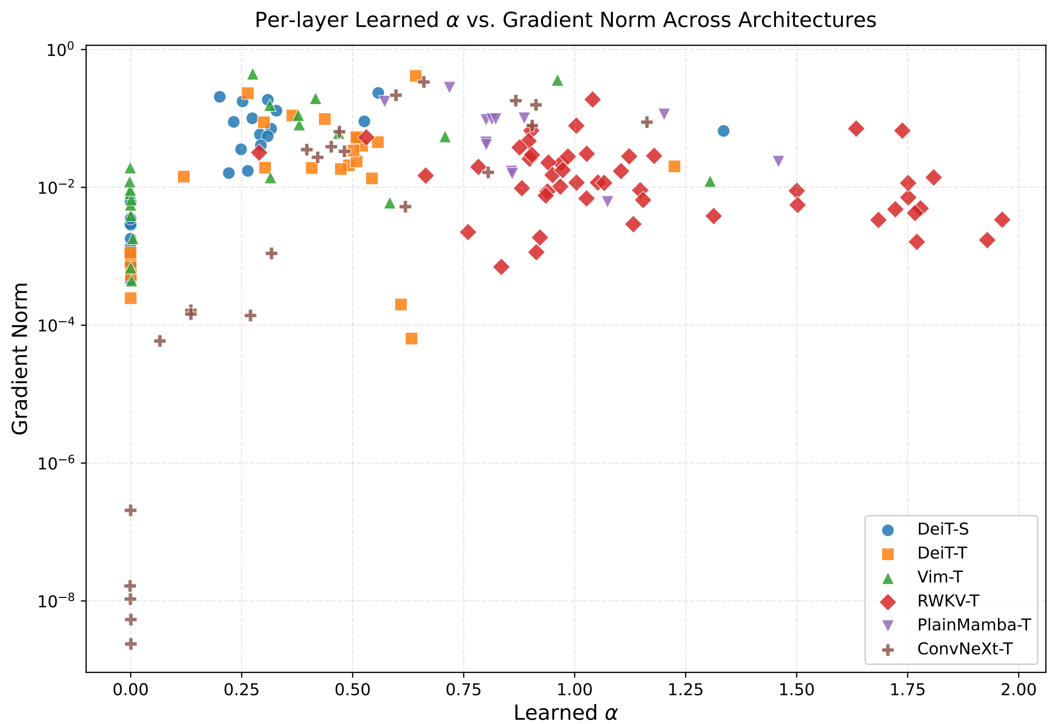

Figure 8: Learned α vs. tanh saturation fraction across architectures (seed = 42). Higher α drives greater saturation, which correlates with effective rank compression (Figure 7).

DyT; in closed-loop architectures with recurrent gradient flow, the substitution may be insufficient as gradient attenuation compounds across sequential state updates. This is consistent with Vim-T's positive Δ ($+3.45$ pp) despite being a closed-loop SSM, and PlainMamba-T's near-zero Δ (-0.41 pp) despite sharing the same topology category. While Vim-T and PlainMamba-T differ in other respects—including absolute accuracy level, model capacity, sensitivity to the DyT scaling parameter α, and PlainMamba-T's learned directional biases [Yang et al., 2024] which may confer implicit position-aware scaling that increases robustness to normalization replacement—the contrast in their scan strategies provides a suggestive, if not conclusive, signal. A controlled experiment varying only scan direction (bidirectional $\rightarrow$ unidirectional in Vim-T) would be needed to confirm this hypothesis; however, such modifications require non-trivial changes to the Vim codebase's scan infrastructure, potentially introducing uncontrolled architectural confounds that would compromise the controlled comparison. As a partial substitute, we ablate PlainMamba-T's directional biases (removing the learned direction-specific scaling while retaining unidirectional scan): the DyT deficit increases modestly from -0.41 pp to -0.87 pp (single seed), consistent with directional biases providing partial implicit normalization but not demonstrating that adding bidirectionality would confer positive Δ.

W.2 Toward Selective Replacement

The preceding analysis treats normalization replacement as an all-or-nothing proposition, yet our results suggest a more nuanced approach. The RWKV-T `key_norm` ablation (Table 20) provides direct evidence: replacing all normalization layers with DyT yields $\Delta = -2.17$ pp, but selectively preserving `key_norm` as LayerNorm while replacing all other layers recovers approximately 35% of the deficit ($\Delta \approx -1.40$ pp, 3-seed mean). A single normalization site—one that stabilizes the query-key dot product—accounts for a disproportionate share of the total degradation.

This finding resonates with the broader literature on position-specific normalization. Query-key normalization has been identified as critical for scaling vision transformers, preventing attention-logit explosion that produces near-one-hot attention weights [Dehghani et al., 2023]. Vision-RWKV [Duan et al., 2024] similarly introduces extra LayerNorm after the WKV attention and Squared ReLU operations specifically to prevent output overflow—an architecture-specific normalization need that a uniform replacement strategy would eliminate. More generally, Peri-LN [Kim et al., 2025] demonstrates that normalization placement itself modulates training dynamics: pre-sublayer and post-sublayer normalizations serve distinct roles in variance control versus gradient stabilization.

These observations motivate a *topology-aware selective replacement* strategy. Rather than replacing all normalization layers uniformly—as in DyT [Zhu et al., 2025], Derf [Chen et al., 2025], or TaperNorm [Kanavalau et al., 2026]—one would classify each normalization site by its functional role (statistical anchoring, gradient stabilization, or activation scaling) and replace only those where the preceding computation already provides equivalent conditioning. Concretely, our results suggest at least two principles: (1) normalization layers that bound dot-product attention (including `key_norm` variants and QK-Norm) should be preserved, as DyT's pointwise $\tanh$ does not replicate the cross-channel standardization these layers provide; and (2) normalization layers following globally-aggregated representations (e.g., post-SA in DeiT) are safer replacement targets, since the full-sequence statistics that LayerNorm computes are already implicit in the activations. The NFNet line of work [Brock et al., 2021] offers a precedent for architecture-specific normalization-free strategies in CNNs, where Scaled Weight Standardization and Adaptive Gradient Clipping jointly compensate for removed batch normalization—a solution that does not transfer to transformers but demonstrates the principle that normalization removal requires mechanism-specific compensation.

Validating this selective strategy at scale—identifying which layers to preserve across diverse architectures and measuring the accuracy–efficiency tradeoff of partial replacement—is a natural next step that our topology-dependent framework is well-positioned to guide.

W.3 DyT Specificity: RMSNorm Evidence

RMSNorm [Zhang and Sennrich, 2019] retains scale normalization but removes mean-centering, providing a controlled intermediate between LayerNorm and DyT (Table 1, RMSNorm column). On DeiT-S, the ordering is LN (58.0%) $<$ RMSNorm ($60.7 \pm 0.3\%$, 3-seed) $<$ DyT (64.9%), with DyT retaining a $+4.2$ pp advantage over RMSNorm—though the primary RMSNorm runs used warmup=5 epochs vs. warmup=0 for LN and DyT. A matched-warmup control (warmup=0, 3-seed) yields $57.8 \pm 1.9\%$ (seeds 42/0/123: 59.5/55.7/58.3%), compared to $60.7 \pm 0.3\%$ with warmup=5. Warmup thus contributes $+2.9$ pp on average ($57.8\% \rightarrow 60.7\%$), substantially larger than the single-seed estimate suggested. Critically, removing warmup amplifies cross-seed standard deviation from 0.3 pp to 1.9 pp—a $\sim 6\times$ increase—revealing that warmup scheduling plays a significant role in stabilizing training for normalization-free alternatives by reducing sensitivity to random initialization. Without warmup, the RMSNorm mean (57.8%) is comparable to LN (58.0%), indicating that the $+2.7$ pp RMSNorm advantage over LN depends on warmup-mediated stabilization; the residual 4.2 pp gap to DyT (64.9%) implicates DyT-specific factors: $\tanh$ gating, learned α, and the complete removal of statistical normalization.

On RWKV-T, the picture inverts: RMSNorm ($72.4 \pm 0.2\%$, 3-seed) and DyT ($72.5 \pm 0.2\%$, 3-seed) are nearly identical ($\Delta = -0.04$ pp, within noise), both degrading ≈ 2 pp relative to LN ($74.6 \pm 0.2\%$). Removing mean-centering alone reproduces the full DyT deficit, indicating that the hybrid architecture's degradation is driven by the loss of centering rather than by DyT's $\tanh$ compression mechanism.

Together, these results demonstrate that the topology-dependent pattern is not an artifact of DyT's specific parameterization: the same normalization modification—removal of mean-centering—produces architecture-dependent outcomes, beneficial in global self-attention when paired with warmup and detrimental in hybrid architectures. DyT amplifies this underlying sensitivity through additional compression, widening the gap where warmup-mediated centering removal is beneficial (DeiT-S: $+2.7 \rightarrow +6.9$ pp) while adding negligible further deficit where it is not (RWKV-T: $-2.1 \rightarrow -2.2$ pp). This architecture-dependent response to a shared normalization change further corroborates the topology-dependent framework of Section 4.

W.4 Cross-Dataset Generalization Discussion

The single-seed Tiny-ImageNet probes (Appendix T) provide preliminary evidence that the topology-dependent pattern extends beyond CIFAR-100. On TIN-200, DeiT-S at native 64×64 resolution shows $\Delta = +1.47$ pp (53.93% vs. 52.46%), and Swin-T at 32×32 shows $\Delta = -9.6$ pp (33.0% vs. 42.6%)—both directionally consistent with CIFAR-100 ($+6.90$ and -17.18 pp, respectively).

The DeiT-S effect attenuation ($+6.90 \rightarrow +1.47$ pp) is accompanied by a more informative signal: DyT compresses the train–validation gap from 25.21 pp (LN) to 7.74 pp—a 3.3× reduction—revealing strong implicit regularization. We interpret the attenuation and regularization as causally linked: $\tanh$ compression limits effective model capacity, which is uniformly beneficial on CIFAR-100 (where 22M parameters exceed task complexity at 32×32) but becomes a binding constraint at 64×64 with $4\times$ the spatial information. The narrower accuracy margin thus reflects DyT's dual role—normalization substitute *and* implicit regularizer—whose net effect is resolution-dependent, since the sign remains positive.

Swin-T's TIN deficit (-9.6 pp) is driven by training accuracy collapse (-33.6 pp vs. LN), consistent with the systematic underfitting observed on CIFAR-100. This cross-dataset replication of windowed-attention failure reinforces the structural incompatibility identified in §5: the deficit is not a dataset-specific artifact but a consequence of windowed attention's interaction with norm-free operation.

Both probes are single-seed; the reported effect sizes are directional evidence, not quantitative estimates of the cross-dataset transfer function. Multi-seed replication at native resolution is needed to determine whether the implicit regularization ceiling is a general property of DyT or specific to the CIFAR-100-to-TIN transfer setting.

Broader Impact

This work is a controlled empirical study of normalization layer alternatives in neural network architectures. We analyze whether Dynamic Tanh (DyT) can replace LayerNorm across architectures with different information-flow topologies (open-loop, closed-loop, hybrid), using standard image classification benchmarks at low resolution.

Potential positive impact. Our findings provide practitioners with topology-aware guidance for when normalization-free techniques can be safely adopted and when they should be avoided. By identifying that DyT effectiveness is topology-dependent rather than universal, we help prevent silent accuracy degradation from blanket adoption of normalization replacement strategies.

Potential negative impact. We do not foresee direct negative societal consequences from this work. The research is a purely architectural analysis that does not introduce new capabilities, datasets, or deployment-facing systems. It does not involve human subjects, personally identifiable information, or dual-use technology.

Computational resources. All experiments were conducted on single NVIDIA RTX 5090 GPUs. Across 117 total experiments (97 with recorded compute), the project consumed approximately **125 GPU-hours** (102.8 from successful experiments, 14.9 from negative-result experiments, and 7.0 from failed or terminated runs). All training was performed at controlled low-resolution settings (32×32 and 64×64), resulting in a modest computational footprint. We estimate the total energy consumption at approximately 50–60 kWh, corresponding to a carbon footprint well under 25 kg CO_2e depending on the electricity grid.

NeurIPS Paper Checklist

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction claim that topology serves as an organizing lens for understanding DyT effectiveness patterns within pre-norm residual architectures, which is directly supported by experiments across ten architectures on CIFAR-100, with cross-dataset validation on Tiny-ImageNet.

2. **Limitations**

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 5.1 discusses dataset scale, single architecture per topology, seed count, absence of formal significance testing, and restriction to vision classification.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (or correct) proof?

Answer: [N/A]

Justification: This is an empirical study with no theoretical results or proofs.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 3 reports all architecture configurations, hyperparameters, training schedules, random seeds, and evaluation protocols. Per-seed results are provided in the Appendix.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: All datasets used (CIFAR-100, Tiny-ImageNet) are publicly available. Code will be released upon publication.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [Yes]

Justification: Section 3 (Method) specifies architectures, optimizer, learning rate schedule, data augmentation, batch size, training epochs, and evaluation protocol. Additional details are in the Appendix.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: We report mean±std across 2–3 seeds for all experiments and provide per-seed results in the Appendix. We acknowledge in Section 5.1 that the small seed count precludes formal significance testing.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: The Appendix reports GPU type and approximate training time per experiment.

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: This work studies normalization techniques in neural network architectures using standard public benchmarks. No ethical concerns arise.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work?

    Answer: [Yes]

    Justification: The Conclusion includes a Broader Impact paragraph noting that our findings serve as practitioner guidance and identifying no anticipated negative societal impacts.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, or scraped datasets)?

    Answer: [N/A]

    Justification: This paper studies normalization in small-scale image classification models. No high-risk models or datasets are released.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: CIFAR-100 (MIT license) and Tiny-ImageNet (derived from ImageNet, publicly available for research) are properly cited. Architecture codebases (DeiT, Vim, Vision-RWKV) are cited and used under their respective open-source licenses.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [N/A]

Justification: No new datasets or pre-trained models are introduced. Code will be released with documentation upon publication.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

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15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the institutional requirements of the authors' research institution) were obtained?

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